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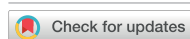
Decoding the chemistry of the FLASH effect: a physicochemical model of dose-rate, pH and oxygen-dependent H_2O_2 production

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Decoding the chemistry of the FLASH effect: a physicochemical model of dose-rate, pH and oxygen-dependent H₂O₂ production

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E-mail: anne-catherine.heuskin@unamur.be, romain.tonneau@unamur.be, rudi.labarbe@iba-group.com
and stephane.lucas@unamur.be**Keywords:** UHDR, FLASH radiotherapy, oxygen effect, hydrogen peroxide, modelling, water radiolysis, radiation chemical yield (G value)

Abstract

Objective. To elucidate the initial chemical mechanisms that may underlie the FLASH effect by developing and validating a unified simulation framework for the radiolysis of pure water. The goal is to create a single model capable of reconciling conflicting experimental and simulation data regarding H₂O₂ production and explaining key radiobiological observations across conventional (CDRs) and ultra-high dose rates (UHDRs) under varied oxygenation levels. **Approach.** An ordinary differential equation-based model was developed to simulate the homogeneous chemistry phase of water radiolysis. The framework incorporates a detailed chemical reaction network and a novel description of acid–base equilibrium, allowing pH to evolve dynamically. A key innovation is the integration of an empirically derived, dose-rate dependent G-value coefficient ($G_F(D_R)$) that anchors the simulation to experimental data. The temporal evolution of key species (H₂O₂, O₂, H₃O⁺, OH[−]) is tracked to investigate the impact of dose rate and oxygen concentration. **Main results.** The model reproduces two key experimental findings relevant to the FLASH effect—previously challenging for simulations: decreased net H₂O₂ production at UHDR under physioxenic conditions. This reduction (vs CDR) aligns with normal tissue sparing, while hypoxic (tumour-like) conditions show comparable H₂O₂ production at UHDR and CDR, consistent with iso-tumour control. These results confirm that H₂O₂ radiochemistry is profoundly influenced by both dose rate and oxygen levels. **Significance.** This work resolves a key discrepancy between previously published simulations and experimental data on UHDR water radiolysis. The model provides a robust, mechanistic foundation linking the physical parameter of dose rate to the distinct chemical environments that likely drive the differential biological outcomes of the FLASH effect. It serves as a powerful new tool for investigating the complex interplay between dose rate, oxygenation, and radiolytic chemistry.

1. Introduction

Water radiolysis is a fundamental physico-chemical process with numerous applications, including radio-sterilization (Maquille *et al* 2008), waste water purification (Guo *et al* 2008), enhancement of the microbiological safety and shelf life of food (Farkas and Mohácsi-Farkas 2011), and in radiotherapy (Chandra *et al* 2021).

Current research in radiotherapy focuses on developing innovative techniques such as FLASH radiotherapy (FLASH-RT) to enhance treatment outcomes and minimize radiation-induced toxicity. FLASH-RT employs ultra-high dose rate (UHDR) irradiation, typically exceeding 100 Gy·s^{−1}, delivered in extremely short pulses (usually <200 ms). This approach has demonstrated a remarkable ability to spare

normal tissue from radiation damage while maintaining equivalent tumour control probability, a phenomenon known as the FLASH effect (Favaudon *et al* 2014). However, the exact reason for this protective mechanism is still under debate, slowing the adoption of FLASH-RT in clinical settings. The emergence of FLASH-RT and the lack of mechanistic explanation raise questions about existing knowledge in standard radiobiology, particularly about the radiolysis of pure water at UHDR. This phenomenon may require a re-evaluation or even a rediscovery of fundamental principles in the field. Understanding these changes could be crucial for advancing both FLASH-RT and our overall grasp of radiation effects.

An initial theory proposed to account for the FLASH effect was a quick reduction in oxygen levels, but this has been tested and dismissed by several studies (Labarbe *et al* 2020, Boscolo *et al* 2021, Cao *et al* 2021, Jansen *et al* 2021, Chappuis *et al* 2023a). UHDR irradiations cause a short-term spike in free radicals and reactive oxygen species (ROS), potentially affecting chemical reactions. Other teams (Montay-Gruel *et al* 2019, Blain *et al* 2022, Kacem *et al* 2022) observed a notable drop in hydrogen peroxide (H_2O_2) production at UHDR compared to conventional dose rates (CDRs). This decline in H_2O_2 is still unexplained. Although it is assumed that radical–radical recombination might occur during UHDR, the specific reactions involved remain unknown.

In parallel, significant efforts have been made to model these phenomena from first principles. Sophisticated Monte Carlo (MC) track-structure simulations, using tools like TOPAS-nBio (Ramos-Mendez *et al* 2020, D-Kondo *et al* 2023), Geant4-DNA (Chappuis *et al* 2023b) and gMicroMC (Lai *et al* 2021), have provided invaluable insights into the ultra-fast physics and chemistry occurring in the initial nanoseconds of water radiolysis, exploring key factors like oxygen consumption and H_2O_2 yields in various media (Shin *et al* 2024, D-Kondo *et al* 2025). These *ab initio* models aim to build a bottom-up understanding by simulating individual particle tracks and their subsequent reactions. While these simulation studies have provided crucial insights into inter-track effects, achieving quantitative agreement with the experimentally observed decrease in H_2O_2 yield at UHDR under physioxic conditions remains an ongoing challenge. This discrepancy is largely attributed not to the simulation methodology in itself, but to the need for more refined chemical input, including specific reaction pathways and rate constants validated for the complex physioxic cellular environment. This highlights a persistent gap between first-principles simulations of early events and macroscopic experimental observations relevant to biological timescales.

The present study takes a different, complementary approach. Rather than modelling the pre-chemical and early chemical stages from scratch, a semi-empirical framework was developed. The model shortcuts the complexity of the first 100 ns by using a novel, dose-rate-dependent G-value model that is directly derived from experimental H_2O_2 measurements from the literature. By anchoring the initial chemical yields to experimental reality, this work aims to bridge the gap between simulation and experiment. The primary goal is not to re-investigate the detailed track physics, but to leverage this simplified, phenomenologically driven input to accurately propagate the homogeneous chemistry to later, biologically relevant times, thereby providing a robust platform to explore the downstream chemical consequences of UHDR irradiation.

In recent years, research groups have explored both experimental and numerical origins of the FLASH effect. On the experimental side, *in vitro* and *in vivo* studies conducted on small animals involve complex living models with many constraints and parameters to consider. This is where the advantages of numerical models come into play as they are not bound by ethical committee approvals or dependent on living materials, enabling the parallel testing of multiple hypotheses. However, these approaches are oversimplified versions of reality. Compromises are necessary, and translating their findings to experimental results remains challenging. Nonetheless, numerical models offer the opportunity to analyse and understand mechanisms that cannot be measured experimentally. In this frame, particle-induced water radiolysis simulations use two primary methods. The first employs MC codes (Watanabe and Saito 2001, Kreipl *et al* 2009, Karamitros *et al* 2014, Boscolo *et al* 2020, Alanazi *et al* 2021, Lai *et al* 2021, Hu *et al* 2022) to examine the physics of particle–water interactions and their chemical consequences. This method provides intricate details at the first moments of irradiation but is limited to brief durations, typically not extending beyond a few microseconds or tens of microseconds.

The second approach seeks to address the time range limitation, allowing computations over a relevant biochemical range (μs –s). Some methods reuse MC simulation outputs focused on very specific parts of the global chemistry (Colliaux *et al* 2011, Meesat *et al* 2012), while others employ conventional chemical reactions, utilizing ordinary differential equations (ODEs) (Labarbe *et al* 2020, Domnanich and Severin 2022, Tan *et al* 2023) to model radiolysis. An advantage of the latter is that it allows the effect of the initial oxygen content to be studied, as O_2 can be modelled without assuming that it is an inexhaustible reservoir, as most MC codes do. Advanced numerical solvers can now process dozens or even hundreds of reactions, allowing them to predict concentration changes over periods up to several hours

(Domnanich and Severin 2022, Tan *et al* 2023). However, a main limitation of these methods is the lack of spatial resolution, as the system is treated as uniform. Because they presuppose uniformity, the early moments of irradiation are not accurately represented and only the homogeneous chemical processes are considered, which typically start between 100 ns and 1 μ s after irradiation.

The aim of this study is therefore to introduce and validate a unified ODE-based model that elucidates the complex chemistry of pure-water radiolysis under CDR and UHDR irradiation across various oxygen levels. The primary novelty of this work is the development of a single simulation framework capable of simultaneously predicting three cornerstone observations relevant to the FLASH effect:

1. The experimentally observed decrease in net H₂O₂ production at UHDR under physioxenic conditions.
2. A marked decrease in H₂O₂ production at UHDR compared with CDR in physioxenic (normal-tissue-like) environments.
3. Comparable H₂O₂ production at both UHDR and CDR under hypoxic (tumour-like) conditions, consistent with the iso tumour control probability seen in FLASH experiments.

To achieve this, the temporal evolution of key species such as H₂O₂, O₂, and the primary drivers of pH dynamics, H₃O⁺ and OH[−], are specifically tracked. Validation of the framework is demonstrated by its ability to reproduce these critical findings, which is made possible by anchoring the simulation with a novel, empirically derived, dose-rate dependent G-value coefficient ($G_F(D_R)$). This work thus provides a new tool to investigate the downstream chemical mechanisms that may support the differential biological response of the FLASH effect.

2. Material and method

2.1. The hydrogen peroxide paradox

Existing research has yielded inconsistent findings regarding the relationship between dose rate and H₂O₂ production in pure water. Notably, simulation studies (Derksen *et al* 2023, Shin *et al* 2024, Zhang *et al* 2024) have contradicted experimental observations (Montay-Gruel *et al* 2019, Blain *et al* 2022, Kacem *et al* 2022, Sunnerberg *et al* 2023, Zhang *et al* 2024) that suggest a decrease in H₂O₂ production under UHDR conditions. Consequently, the impact of UHDR on H₂O₂ yield relative to conventional irradiation remains unclear.

A significant challenge in reconciling these findings is that experimental quantification of H₂O₂ in irradiated water requires the addition of chemical probes. This approach introduces a confounding factor: the probes and their required co-factors (e.g. enzymes) are added to the system, meaning the chemistry is no longer that of ‘pure water’. These added compounds may interact with transient radiolytic species, which could alter the chemical kinetics and temporal evolution that they are intended to measure. The two most common chemical probes used to measure the production of H₂O₂ are Amplex Red and Ghormley triiodide. The Amplex Red assay, in which 10-acetyl-3,7-dihydroxyphenoxazine reacts 1:1 with H₂O₂ in the presence of horseradish peroxidase (HRP) enzyme, produces the fluorescent compound resorufin. In the Ghormley Triiodide method H₂O₂ reacts with iodide ions (I[−]) to form triiodide ions (I₃[−]), which are quantified via light absorbance. Given that these two assays rely on fundamentally different chemical pathways, the probe choice may be a major, and perhaps underappreciated, source of the variability in reported H₂O₂ yields.

The studies of the last 50 years, focusing on H₂O₂ production in pure water upon irradiation have been compiled in table 1. Indications about radiation types, linear energy transfer (LET), oxygen level in water prior to irradiation as well as the H₂O₂ probing method used are listed. The coloured squares serve as legend for figure 1.

In figure 1, all data available in the studies reported in table 1 are translated in G values units i.e. expressed as #/100 eV (1#/100 eV $\cong$ 0.104 μ M $\cdot$ Gy^{−1}). All data were consolidated to utilize a unified mean dose rate ($\overline{D_R}$) formula using the same beam structure conventions as (Tobias Bohlen *et al* 2024):

$$\overline{D_R} = \frac{D_{\text{beam}}}{T_{\text{beam}}} = \frac{\#_{\text{pulse}} \cdot D_{\text{pp}}}{(\#_{\text{pulse}} - 1) \cdot \Delta t_{\text{pulse}} + \#_{\text{pulse}} \cdot t_{\text{pulse}}} \quad (1)$$

where D_{beam} represents the total dose (Gy), T_{beam} is the total beam-on time (s), $\#_{\text{pulse}}$ denotes the number of pulses, D_{pp} is the dose per pulse (Gy), t_{pulse} the pulse duration (s) and Δt_{pulse} the time duration in between two consecutive pulses.

From this figure, the two experimental protocols for quantifying H₂O₂ form two clusters with noticeable differences. All Amplex Red based data points have 58% to 263% higher yields than the others. The

Table 1. Published experimental works reporting H₂O₂ G-values in pure water.

Reference		Radiation type	LET (keV·μm ⁻¹)	H ₂ O ₂ probe method	Oxygen (% pO ₂)
Kacem et al (2022)	■	Electrons	0.6	Amplex Red	4%
		Protons	0.4		
Montay-Gruel et al (2019)	■	Electrons	0.6	Amplex Red	4%
Zhang et al (2024)	■	Electrons	0.6	Amplex UltraRed	0.1% 21%
Blain et al (2022)	■	Protons	1.17	Ghormley triiodide	Aerated water
Sehested et al (1968)	■	Electrons	0.4	Home made	Aerated water
Roth et al (2011)	■	Gamma	0.21	Ghormley triiodide	Aerated water
Stefanic et al (2002)	■	Gamma	0.21	Ghormley triiodide	Aerated water
Rafi et al (1965)	■	Gamma	0.21	Ghormley triiodide	Aerated water
Trupin et al (2002)	■	Gamma	0.53	Chemiluminescence	Deaerated water
		Protons	2.5	calibrated with	
		Carbon	30	Ghormley triiodide	

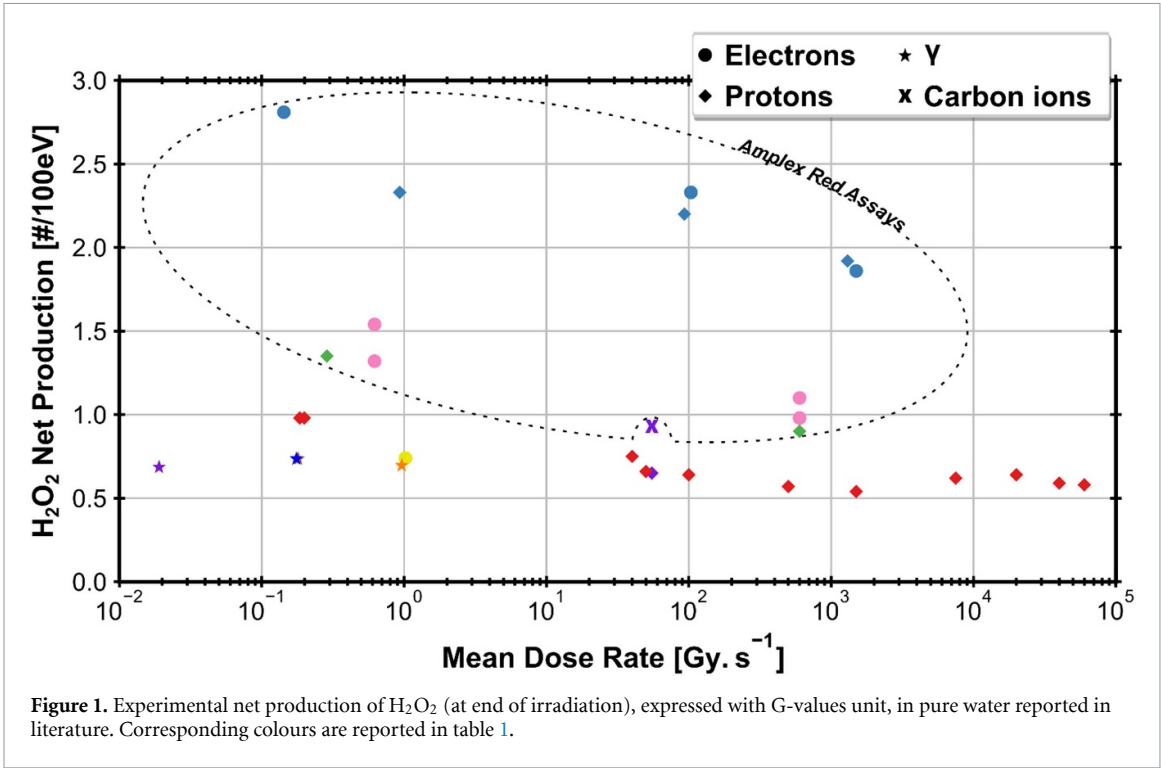


Figure 1. Experimental net production of H₂O₂ (at end of irradiation), expressed with G-values unit, in pure water reported in literature. Corresponding colours are reported in table 1.

only exception is the data point corresponding to carbon ion beam at 55 Gy·s⁻¹. However, this data point is the only one from the data set with a high LET of 30 keV·μm⁻¹, much higher than the others (≤2.5 keV·μm⁻¹). At such high LET, intertrack effects, which are outside the low-LET scope of this study, are known to alter radical recombination kinetics and reduce the net H₂O₂ yield, thus explaining its deviation from the other measurements (Ramos-Mendez et al 2020). Based on this observation, the present study focused on data measured without Amplex Red and is restricted to the one of Blain et al (2022). Indeed, the latter is the only study to have experimentally measured the production of H₂O₂ over a broad range of mean dose rates. Moreover, despite using a cyclotron (inherently pulsed with MHz micro-structure), the authors have performed all UHDR irradiations with a single macro pulse. Finally, gamma-based data are not included as the LET used by Blain et al (2022) is one order of magnitude larger.

2.2. The simulation software

Labarbe et al (2020) demonstrated the feasibility of integrating published G-values from MC codes to set up initial conditions that are subsequently processed by another approach using an ODE solver. The methodology applied in this paper uses published data from MC code (Boscolo et al 2020) as input for a custom Python code that solves a system of ODEs.

The time-dependent differential equations describing the evolution of n chemical species x_i can be written as:

$$\frac{dx_i}{dt} = f_{P,i}(x_1, x_2, \dots, x_n) - f_{C,i}(x_1, x_2, \dots, x_n) + R_i(t) \quad (2)$$

where $i = 1, \dots, n$; $f_{P,i}(x_1, x_2, \dots, x_n)$ is the production term ($\mu\text{M}\cdot\text{s}^{-1}$) of the species i ; $f_{C,i}(x_1, x_2, \dots, x_n)$ is the consumption term ($\mu\text{M}\cdot\text{s}^{-1}$) of the species i and $R_i(t)$ is the radiolytic production term defined by:

$$R_i(t) = G_{\text{CDR},i}(t = 100\text{ns}) \cdot G_F \left(\frac{dD(t)}{dt} \right) \cdot D \cdot \frac{\rho}{e \cdot 100 \cdot N_A} \quad (3)$$

where $G_{\text{CDR},i}(t = 100\text{ns})$ is the G -value ($\#/100 \text{ eV}$), at 100 ns, of the species i from MC simulations (literature) for CDR; $\frac{dD(t)}{dt}$ is the instantaneous dose rate ($\text{Gy}\cdot\text{s}^{-1}$); $G_F \left(\frac{dD(t)}{dt} \right)$ is the dimensionless dose rate dependent coefficient (' G -factor') introduced in this study; ρ is the density ($\text{kg}\cdot\text{l}^{-1}$) of the considered medium (water in this study); D the deposited dose (Gy) during dt ; e the elementary charge and N_A the Avogadro's constant.

Numerous methods are available for solving such set of differential equations. However, in the specific case of chemical kinetics, this problem is defined as a stiff system. The main issue of numerical methods for ODE's stiff system solution is the numerical stability. Fortunately, there are efficient ODE solvers suitable for stiff problems. One of them is the LSODA solver (Petzold 1983). LSODA differs from the other integrators in that it can switch automatically between stiff and non-stiff methods. There is then no need to assess the stiffness in advance as the solver will automatically switch to the appropriate method. LSODA always starts with the non-stiff method. When stiffness is detected in the solution, the solver backtracks to switch the resolution method and adapt the time step so that quick variations of the solution can be resolved. This method is particularly well suited for the chemical kinetics of water radiolysis, as one can expect discontinuities in the chemical kinetics when beam is switched ON and OFF. This solver is made available in the Python environment via the scipy module (Virtanen *et al* 2020). In this work, the radiolysis model, developed with Python 11.9, wraps the LSODA solver and builds the chemical environment out of a TOML configuration file listing all reactions detailing the radiolysis (see table 2). The kinetics of the homogeneous chemistry phase related to pure water radiolysis is described by the set of the elementary reactions given in table 2.

A schematic description of the developed code is shown on figure 2.

The simulation code parses the input parameters (beam definition, chemical reactions involved, initial pH and species concentrations) from an input file (TOML syntax) and constructs the chemical and physical environment. The beam is defined by the LET, total dose and instantaneous dose rate. A pulsed beam can be created providing pulse width and period or frequency. The relevant chemical species are automatically identified and included within the simulation environment based on (i) the definition of the chemical reactions and (ii) the initial concentrations provided in the input file. Petersen-like matrices representing the chemistry network described in the input file are automatically created. These matrices are describing reaction constants, reactants, and products of each reaction as well as their respective stoichiometry. Equilibrium reactions are treated as two elementary reactions (forward and backward). Initial concentrations are assumed equal to zero if not specified otherwise in the input file. Only the concentration of water is hardcoded as its concentration is assumed not to be affected by irradiation. Initial concentration of H_3O^+ and OH^- are directly computed from the initial pH value (assumed neutral if not specified). These main simulation parameters are listed in table 3. No reaction is hard coded to give the end user the freedom to vary/include/omit any reaction parameter. This simulation package is built upon the work of Labarbe *et al* (2020), i.e. the model is extended with a set of chemical reactions relevant to the acid-base chemistry and the effect of radiations is accounted for by using G -values from (Boscolo *et al* 2020). Water radiolysis products are formed continuously, according to these G -values, during the period of irradiation until the desired dose or pulse width is reached. These values are available as a function of the LET in (Boscolo *et al* 2020) and correspond to 1 μs of simulation, starting from the physical phase. In the absence of oxygen, the G -values already achieve equilibrium by 100 ns. Therefore, the simulations are started at 100 ns with these values, enabling to tackle the effect of oxygen on the primary products at the early times of the homogeneous chemistry phase. The G -values as function of the LET of the following species: e_{aq}^- , $\text{HO}\cdot$, H_2O_2 , $\text{H}\cdot$ and H_2 are included. About the G -value of H_3O^+ , it is assumed equal to the one of e_{aq}^- to guarantee charge conservation. To simulate the impact

Table 2. List of all chemical reactions included in the model. Reaction rates are mainly from (A) (Domnanich and Severin 2022), (B) are rates from (Joseph et al 2008), (C) are rates from (Labarbe et al 2020) and (D) are rates from (Buxton et al 1988). All reaction rate constants correspond to a water temperature of 25 °C.

#	Reaction	Rate constant	#	Reaction	Rate constant	
1	$2e_{aq}^- + 2H_2O \rightarrow H_2 + 2OH^-$	5.50×10^9	(B)	38	$HO_2 + O_3^- \rightarrow 2O_2 + OH^-$	6.00×10^9 (A)
2	$e_{aq}^- + H^\cdot + H_2O \rightarrow H_2 + OH^-$	2.76×10^{10}	(A)	39	$HO_2 + O_3 \rightarrow HO_3 + O_2$	5.00×10^8 (A)
3	$e_{aq}^- + OH^\cdot \rightarrow OH^-$	3.00×10^{10}	(B)	40	$O_2^- + O^- + H_2O \rightarrow O_2 + 2OH^-$	6.00×10^8 (A)
4	$e_{aq}^- + O^- + H_2O \rightarrow 2OH^-$	2.20×10^{10}	(A)	41	$2O_2^- + 2H_2O \rightarrow H_2O_2 + O_2 + 2OH^-$	0.3 (A)
5	$e_{aq}^- + H_2O_2 \rightarrow OH^\cdot + OH^-$	1.10×10^{10}	(C)	42	$O_2^- + H_2O_2 \rightarrow OH^\cdot + O_2 + OH^-$	0.13 (A)
6	$e_{aq}^- + HO_2^- \rightarrow O^- + OH^-$	3.50×10^9	(A)	43	$O_2^- + HO_2^\cdot \rightarrow O^- + O_2 + OH^-$	0.13 (A)
7	$e_{aq}^- + O_2 \rightarrow O_2^-$	1.90×10^{10}	(C)	44	$O_2^- + O_3^- + H_2O \rightarrow 2O_2 + 2OH^-$	1.00×10^4 (A)
8	$e_{aq}^- + O_2^- + H_2O \rightarrow HO_2^- + OH^-$	1.30×10^{10}	(A)	45	$O_2^- + O_3 \rightarrow O_3^- + O_2$	1.50×10^9 (A)
9	$e_{aq}^- + HO_2^\cdot \rightarrow HO_2^-$	1.30×10^{10}	(A)	46	$2O^- + H_2O \rightarrow HO_2^- + OH^-$	1.00×10^9 (A)
10	$e_{aq}^- + O_3^- + H_2O \rightarrow O_2 + 2OH^-$	1.60×10^{10}	(A)	47	$O^- + O_2 \rightarrow O_3^-$	3.75×10^9 (A)
11	$e_{aq}^- + O_3 \rightarrow O_3^-$	3.60×10^{10}	(A)	48	$O^- + H_2 \rightarrow H^\cdot + OH^-$	1.28×10^8 (A)
12	$OH^\cdot + H_2O_2 \rightarrow HO_2^\cdot + H_2O$	2.20×10^7	(D)	49	$O^- + H_2O_2 \rightarrow O_2^- + H_2O$	5.00×10^8 (A)
13	$OH^\cdot + O^- \rightarrow HO_2^-$	2.50×10^{10}	(A)	50	$O^- + HO_2^- \rightarrow O_2^- + OH^-$	7.86×10^8 (A)
14	$OH^\cdot + HO_2^- \rightarrow HO_2^\cdot + OH^-$	7.50×10^9	(C)	51	$O^- + O_3^- \rightarrow 2O_2^-$	7.00×10^8 (A)
15	$OH^\cdot + O_3^- \rightarrow O_3 + OH^-$	2.60×10^9	(A)	52	$O^- + O_3 \rightarrow O_2^- + O_2$	5.00×10^9 (A)
16	$OH^\cdot + O_3^- + H_2O \rightarrow 2O_2^- + H_3O^+$	6.00×10^9	(A)	53	$O_3^- \rightarrow O_2 + O^-$	2.62×10^3 (A)
17	$OH^\cdot + O_3 \rightarrow HO_2^\cdot + O_2$	1.10×10^8	(A)	54	$O_3^- + H_3O^+ \rightarrow O_2 + OH^\cdot + H_2O$	9.00×10^{10} (A)
18	$OH^\cdot + H^\cdot \rightarrow H_2O$	1.09×10^{10}	(A)	55	$HO_3 \rightarrow O_2 + OH^\cdot$	1.10×10^5 (A)
19	$OH^\cdot + OH^\cdot \rightarrow H_2O_2$	5.50×10^9	(C)	Acid Base related reactions		
20	$OH^\cdot + H_2 \rightarrow H^\cdot + H_2O$	3.95×10^7	(A)	56	$HO_2 + OH^- \rightarrow O_2^- + H_2O$	1.33×10^{10} (A)
21	$OH^\cdot + HO_2^\cdot \rightarrow O_2 + H_2O$	8.84×10^9	(A)	57	$O_2^- + H_2O \rightarrow HO_2^\cdot + OH^-$	0.155 (A)
22	$OH^\cdot + O_2^- \rightarrow OH^- + O_2$	8.00×10^9	(B)	58	$2H_2O \rightarrow H_3O^+ + OH^-$	2.11×10^{-5} (A)
23	$H^\cdot + H_2O \rightarrow H_2 + OH^\cdot$	11	(B)	59	$H_3O^+ + OH^- \rightarrow 2H_2O$	1.18×10^{11} (A)
24	$2H^\cdot \rightarrow H_2$	7.52×10^9	(B)	60	$OH^\cdot + H_2O \rightarrow H_3O^+ + O^-$	9.43×10^{-2} (A)
25	$H^\cdot + H_2O_2 \rightarrow OH^\cdot + H_2O$	9.00×10^7	(B)	61	$H_3O^+ + O^- \rightarrow OH^\cdot + H_2O$	5.02×10^{10} (A)
26	$H^\cdot + O^- \rightarrow OH^-$	1.00×10^{10}	(A)	62	$H_2O_2 + H_2O \rightarrow H_3O^+ + HO_2^-$	9.43×10^{-2} (A)
27	$H^\cdot + HO_2^- \rightarrow OH^\cdot + OH^-$	9.00×10^7	(A)	63	$H_3O^+ + HO_2^- \rightarrow H_2O_2 + H_2O$	5.02×10^{10} (A)
28	$H^\cdot + O_3^- \rightarrow OH^- + O_2$	1.00×10^{10}	(A)	64	$HO_2^\cdot + H_2O \rightarrow H_3O^+ + O_2^-$	7.73×10^5 (A)
29	$H^\cdot + O_2^- \rightarrow HO_2^-$	1.14×10^{10}	(A)	65	$H_3O^+ + O_2^- \rightarrow HO_2^\cdot + H_2O$	5.02×10^{10} (A)
30	$H^\cdot + O_3 \rightarrow HO_3$	3.80×10^{10}	(A)	66	$e_{aq}^- + H_2O \rightarrow H^\cdot + OH^-$	19 (D)
31	$H^\cdot + O_2 \rightarrow HO_2^\cdot$	1.31×10^{10}	(A)	67	$H^\cdot + OH^- \rightarrow e_{aq}^- + H_2O$	2.20×10^7 (D)
32	$H^\cdot + HO_2^\cdot \rightarrow H_2O_2$	1.14×10^{10}	(A)	68	$H_2O_2 + OH^- \rightarrow HO_2^- + H_2O$	1.33×10^{10} (A)
33	$HO_2^\cdot + O_2^- \rightarrow HO_2^- + O_2$	8.00×10^7	(A)	69	$HO_2^- + H_2O \rightarrow H_2O_2 + OH^-$	1.27×10^6 (A)
34	$2HO_2^\cdot \rightarrow H_2O_2 + O_2$	8.40×10^5	(A)	70	$e_{aq}^- + H_3O^+ \rightarrow H^\cdot + H_2O$	2.09×10^{10} (A)
35	$HO_2^\cdot + O^- \rightarrow O_2 + OH^-$	6.00×10^9	(A)	71	$H^\cdot + H_2O \rightarrow e_{aq}^- + H_3O^+$	5.83 (A)
36	$HO_2^\cdot + H_2O_2 \rightarrow OH^\cdot + O_2 + H_2O$	0.50	(A)	72	$OH^\cdot + OH^- \rightarrow O^- + H_2O$	1.33×10^{10} (A)
37	$HO_2^\cdot + HO_2^- \rightarrow OH^\cdot + O_2 + OH^-$	0.50	(A)	73	$O^- + H_2O \rightarrow OH^\cdot + OH^-$	1.27×10^6 (A)

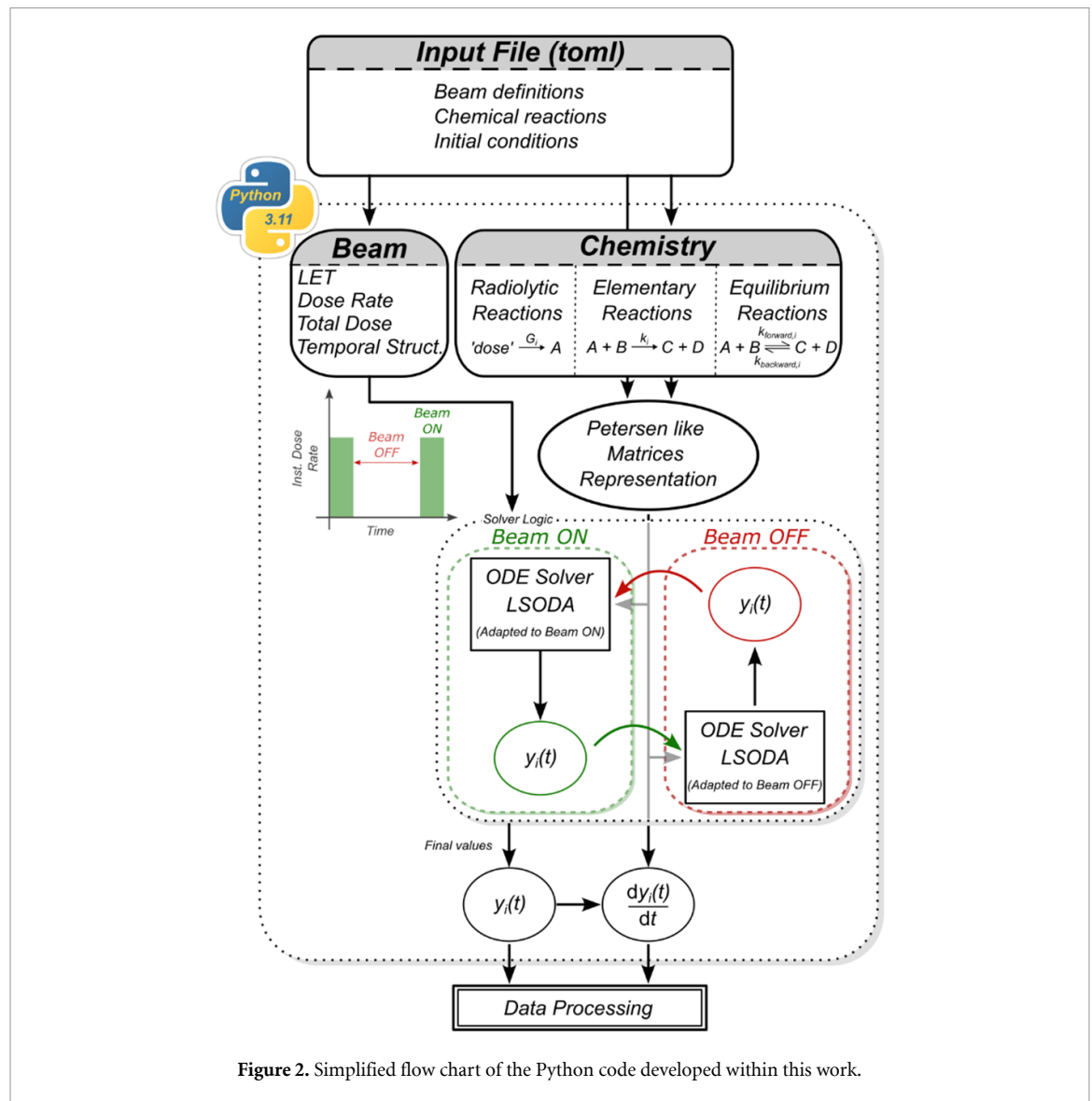


Figure 2. Simplified flow chart of the Python code developed within this work.

of dose rate dependent G-Values, a coefficient ($G_F(D_R)$) is added and defined by:

$$G(\text{species}, \text{LET}, D_R, 100\text{ns}) = G_{\text{CDR}}(\text{species}, \text{LET}, 100\text{ns}) \times G_F(D_R) \quad (4)$$

where G_{CDR} are the initial G-values as defined in (Boscolo *et al* 2020). Unfortunately, the scientific literature contains very little data about the impact of UHDR on G-values. Therefore, the present study aims at determining the relationship $G_F(D_R)$ and assuming its validity for all species, i.e. for a single dose rate G_F is the same for all species. A more detailed approach should also consider the variation of G_F with the species, $G_F(\text{species}, D_R)$. However, due to the lack of both experimental and simulation data the present study assumes G_F as species-independent as it would require tackling the physical and heterogeneous phases of water radiolysis. The formalism developed in this code cannot be extended to the earlier phases of water radiolysis.

At the end of the irradiation time, i.e. when the desired dose is delivered, the solver stops and transfers the calculated species concentrations to another LSODA solver with finetuned parameters adapted to 'OFF' beam. This decomposition into several steps enables to speed up the calculations and lower memory consumption. Indeed, during the relaxation time (i.e. beam 'OFF') the time step resolution can be increased without adding errors to the result. Therefore, if the defined beam is pulsed, the code will automatically divide the simulation time domain into multiple parts corresponding to 'ON' and 'OFF' beam times.

The outcome of the simulation is the time evolution, $x_i(t)$, of each species, i , involved in the network of chemical reactions. From this, it is possible to re-use the same set of matrices as the one employed by

Table 3. Parameter values for the simulations performed in this work.

Simulation parameters		Value (s)
Initial time (s)		1×10^{-7}
Final time (s)		500
Beam structure		Single pulse
Pulse width		Adjusted according to targeted dose rate and total dose
Total dose (Gy)		30
Maximum time step during beam ON		Adjusted depending on pulse width to ensure a minimum of 500 iterations over a pulse duration
Henry's constant to convert pO ₂ to [O ₂]		$1.3 \times 10^3 \mu\text{M}\cdot\text{atm}^{-1}$
Maximum time step during beam OFF (s)		0.1
Initial time step (s)		1×10^{-15}
LSODA absolute tolerance (μM)		1×10^{-12}
LSODA relative tolerance		1×10^{-10}
Temperature ($^{\circ}\text{C}$)		25
Initial concentrations (M)	H ₂ O	55.5
	H ₃ O ⁺	1×10^{-7}
	OH [−]	1×10^{-7}
	O ₂	Variable
	All others	0.0
LET ($\text{keV}\cdot\mu\text{m}^{-1}$)		1.17
G values at 100 ns (G_{CDR}) for LET of $1.17 \text{ keV}\cdot\mu\text{m}^{-1}$ [molecules/100 eV]	e _{aq} [−]	2.49
	H ₃ O ⁺	2.49
	HO [•]	2.62
	H [•]	0.59
	H ₂	0.31
	H ₂ O ₂	0.54

the LSODA solver to evaluate, for each time step, what were the reaction rates, $\frac{dx_i}{dt}(t)$ and the production (F_P) and consumption rates (F_C). The code is available upon request.

Unless otherwise specified, the modelling presented in this study follows the configuration detailed in table 3.

3. Results

3.1. Determining the $G_F(D_R)$ relationship

This section details the systematic procedure used to determine the mathematical relationship for $G_F(D_R)$ by calibrating the simulations against published experimental data from (Blain *et al* 2022) measured in ultra-pure water at 21% pO₂. The overall strategy involves using the experimentally measured production of hydrogen peroxide ($G_{\text{H}_2\text{O}_2}^{\text{exp}}$), a key radiolysis product, as a benchmark. A series of simulations is performed to establish a clear relationship between the input G_F factor and the resulting simulated H₂O₂ yield ($G_{\text{H}_2\text{O}_2}^{\text{sim}}$). By matching this simulated yield to the experimental yield, the most accurate G_F value for each dose rate can be deduced.

This calibration was accomplished through the following four-steps process:

1. **Establishing a model calibration curve for each beam structure:** For each distinct beam structure listed in table 4, a unique relationship between the input parameter G_F and the model's output $G_{\text{H}_2\text{O}_2}^{\text{sim}}$ is established. The conditions, detailed in table 4, include various beam structures for both CDR (pulsed beams) and UHDR (continuous beams), each corresponding to a different mean dose rate. For each beam structure:
 - a. **Parameter scan:** A comprehensive set of simulations is performed where the input G_F coefficient was systematically varied from 0.1 to 2.0 in increments of 0.1.
 - b. **Accurate $G_{\text{H}_2\text{O}_2}^{\text{sim}}$ calculation:** For each of the tested G_F values, the corresponding $G_{\text{H}_2\text{O}_2}^{\text{sim}}$ is calculated. To ensure accuracy and avoid reliance on a single data point, simulations are run at multiple total doses (from 0.1 Gy to 30 Gy). The final [H₂O₂] is plotted as a function of total dose, and a linear regression is applied. The slope of this line provides a robust measure of $G_{\text{H}_2\text{O}_2}^{\text{sim}}$ for that specific G_F value and beam structure.

Table 4. List of the beam structures used in this study for the determination of $G_F(D_R)$.

Mode	Period(ms)	#Pulses	Pulse width (ms)	Inst. dose rate ($\text{Gy}\cdot\text{s}^{-1}$)	Mean dose rate ($\text{Gy}\cdot\text{s}^{-1}$)
CDR	10	15 000	0.8	2.5	0.2
	20	7500	0.1	40	0.2
UHDR	N.A.	1	750 (30 Gy)	40	40
	N.A.	1	600 (30 Gy)	50	50
	N.A.	1	300 (30 Gy)	100	100
	N.A.	1	60 (30 Gy)	500	500
	N.A.	1	20 (30 Gy)	1.5×10^3	$1.5 \cdot 10^3$
	N.A.	1	4 (30 Gy)	7.5×10^3	$7.5 \cdot 10^3$
	N.A.	1	1.5 (30 Gy)	20×10^3	$20 \cdot 10^3$
	N.A.	1	0.75 (30 Gy)	40×10^3	$40 \cdot 10^3$
	N.A.	1	0.5 (30 Gy)	60×10^3	$60 \cdot 10^3$

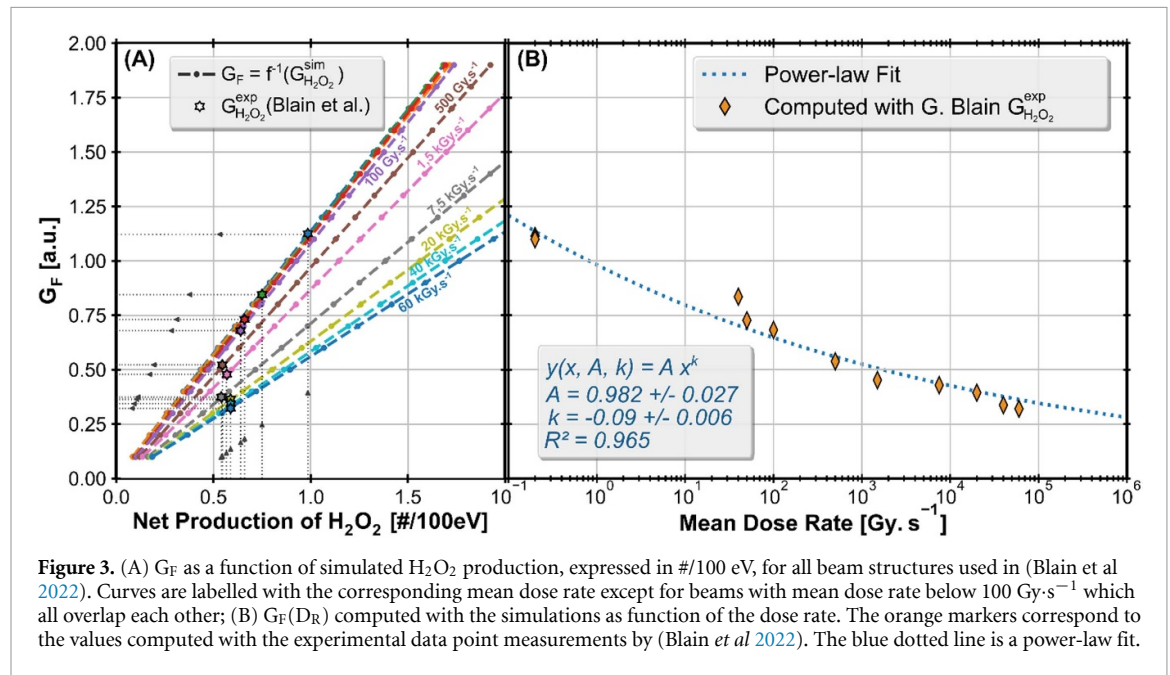


Figure 3. (A) G_F as a function of simulated H_2O_2 production, expressed in #/100 eV, for all beam structures used in (Blain et al 2022). Curves are labelled with the corresponding mean dose rate except for beams with mean dose rate below $100 \text{ Gy}\cdot\text{s}^{-1}$ which all overlap each other; (B) $G_F(D_R)$ computed with the simulations as function of the dose rate. The orange markers correspond to the values computed with the experimental data point measurements by (Blain et al 2022). The blue dotted line is a power-law fit.

- c. **Generating the $G_{\text{H}_2\text{O}_2}^{\text{sim}} = f(G_F)$ relationship:** By repeating this process for all G_F values, a series of $(G_F, G_{\text{H}_2\text{O}_2}^{\text{sim}})$ data points is generated. For each beam structure, these points exhibit a strong linear relationship (see figure 3(A)), which is fitted to create a calibration function of the form $G_{\text{H}_2\text{O}_2}^{\text{sim}} = f(G_F)$.
2. **Calibrating G_F values with experimental data:** With the calibration function established for a given beam structure, the corresponding experimental value of $G_{\text{H}_2\text{O}_2}^{\text{exp}}$ as published by (Blain et al 2022) is selected so that $G_{\text{H}_2\text{O}_2}^{\text{exp}} = G_{\text{H}_2\text{O}_2}^{\text{sim}}$. Applying the inverse of the calibration function, $G_F = f^{-1}(G_{\text{H}_2\text{O}_2}^{\text{exp}})$, determines the unique G_F value required for the model to precisely reproduce the experimental finding at that specific beam structure.
3. **Finalizing the $G_F(D_R)$ relationship:** This entire procedure is carried out for every beam structure and dose rate investigated by (Blain et al 2022). The resulting set of (D_R, G_F) pairs corresponds to the orange markers plotted in figure 3(B). To create a continuous function for use in subsequent simulations, these data points are fitted with a power-law, resulting in the final $G_F(D_R)$ relationship shown by the blue dotted line in figure 3(B).

The fit of the evolution of the $G_F(D_R)$ coefficient with the dose rate is implemented in the simulation code so that the $G_{\text{CDR}}(\text{species}, \text{LET}, 100 \text{ ns})$ is dynamically adapted based on the dose rate of any given irradiation scenario.

Table 5. List of dose rates, along with associated colours used throughout this work, and related pulse width used for simulations.

	Dose rate ($\text{Gy}\cdot\text{s}^{-1}$)		Pulse width
CDR		0.1	300 s
		1.0	30 s
		10	3 s
		50	0.6 s
UHDR		10^2	0.3 s
		10^3	30 ms
		10^4	3 ms
		10^5	0.3 ms
		10^6	30 μs
		5×10^6	6 μs
		10^7	3 μs

3.2. Dose rate dependent simulations

Using the G -value definition of equation (4) with the G_F as computed in figure 3, new simulations are performed with a single pulse of 30 Gy, a LET of $1.17 \text{ keV } \mu\text{m}^{-1}$, with the pulse width adjusted to match the range of simulated dose rates shown in table 5.

To date, the limit between CDR and UHDR is still not clearly defined. This work uses the same threshold as in the work of Pratx (Pratx and Kapp 2019) i.e. UHDR regime starts from a dose rate of $100 \text{ Gy}\cdot\text{s}^{-1}$. Indeed, most of the studies highlighting the FLASH effect and reviewed in (Romano *et al* 2022) have used an average dose rate higher than $100 \text{ Gy}\cdot\text{s}^{-1}$. For the forthcoming analysis of the simulation results, an average physioxic oxygen level of 5% (65 μM) is assumed for healthy cells, as reported by (McKeown 2014). This value lies within the broader physioxic range of 4%–7.5% pO_2 (52.0–97.5 μM), while any condition below 4% is considered hypoxic.

3.2.1. Oxygen

At CDR (figures 4(A) and (C)): For both oxygen levels, $[\text{O}_2]$ decreases by $0.8\text{--}1.2 \mu\text{mol}\cdot\text{l}^{-1}$ upon delivery of 30 Gy. When the beam is switched off, the oxygen level stays steady.

At UHDR (figures 4(B) and (D)): The final oxygen concentration (i.e. at end of simulation time) is in the same range as CDR irradiations. However, for the highest dose rates ($\geq 10^4 \text{ Gy}\cdot\text{s}^{-1}$), $[\text{O}_2]$ decrease is more important at the end of the beam ON time and is followed by a recovery phase of the oxygen content at beam OFF. The recovery phase is split into two consecutive phases. The O_2 net production is first attributed to R_{22} ($\text{OH}\cdot + \text{O}_2^- \rightarrow \text{OH}^- + \text{O}_2$, for times $\leq 1 \text{ ms}$ after 30 Gy delivery), because it involves $\text{HO}\cdot$ which is available during beam ON and reaches higher concentrations in UHDR. The second increase (after 1 ms) of O_2 concentration is related to R_{33} ($\text{HO}_2 + \text{O}_2^- \rightarrow \text{HO}_2^- + \text{O}_2$). This last reaction can indeed sustain a higher reaction rate after beam ON because it does not involve primary products of irradiation as reactants. Therefore, the reactants have a longer lifetime.

To provide a better overview of the oxygen depletion across several initial pO_2 values, figure 5 plots the final (i.e. at 500 s) oxygen depletion ($\Delta[\text{O}_2]$) as a function of both the initial pO_2 and the dose rate.

To further examine and understand $\Delta[\text{O}_2]$, the time evolution of reaction rates producing ($F_P(t)$) or consuming O_2 ($F_C(t)$) are analysed from simulation data. $F_P(t)$ and $F_C(t)$ are then numerically integrated over the time span of the simulation (i.e. from 0 to 500 s) to quantify the gross production and consumption. These data are presented in figure 6.

In figure 6, the small horizontal black lines indicate the net O_2 depletion. At CDR and 0% pO_2 , the net depletion of O_2 is 0 μM but still, 7.5 μM is produced and consumed over the simulation time. At higher dose rates both production and consumption are reduced.

For CDR ($< 100 \text{ Gy}\cdot\text{s}^{-1}$): Initially ($\text{pO}_2 < 0.5\%$), O_2 production and consumption yields have similar values, resulting to an almost zero depletion. As pO_2 increases, both yields increase with production yield (dark colours) experiencing a less pronounced increase. This differential decrease in consumption and production yields, leads to an overall increase in net depletion, saturating at $0.8\text{--}1 \mu\text{M}$

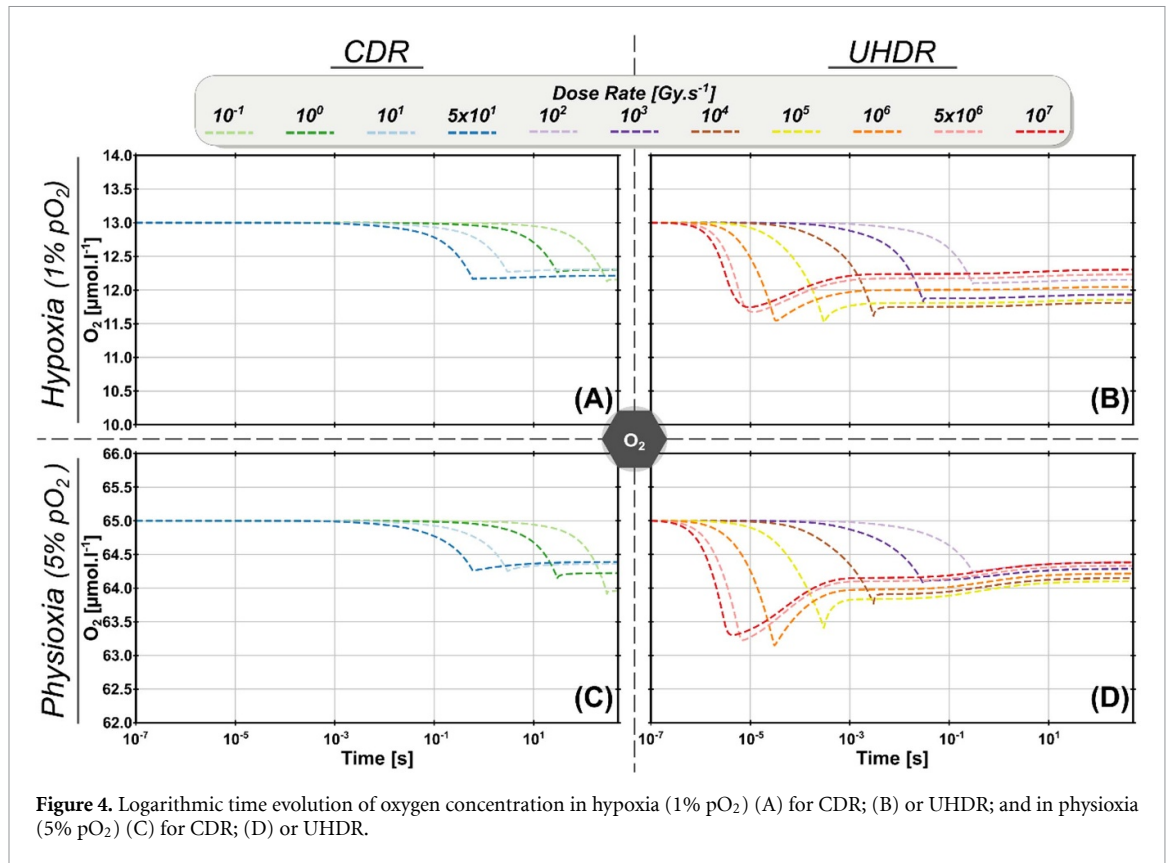


Figure 4. Logarithmic time evolution of oxygen concentration in hypoxia (1% pO₂) (A) for CDR; (B) or UHDR; and in physioxia (5% pO₂) (C) for CDR; (D) or UHDR.

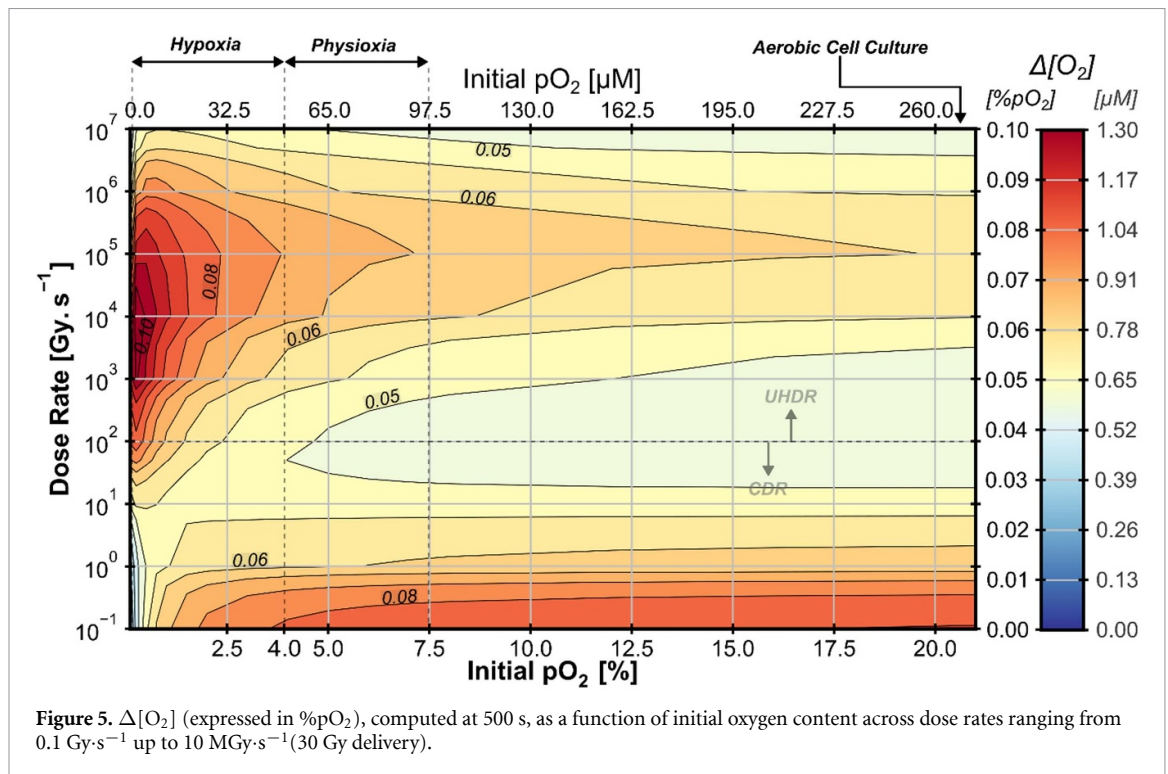
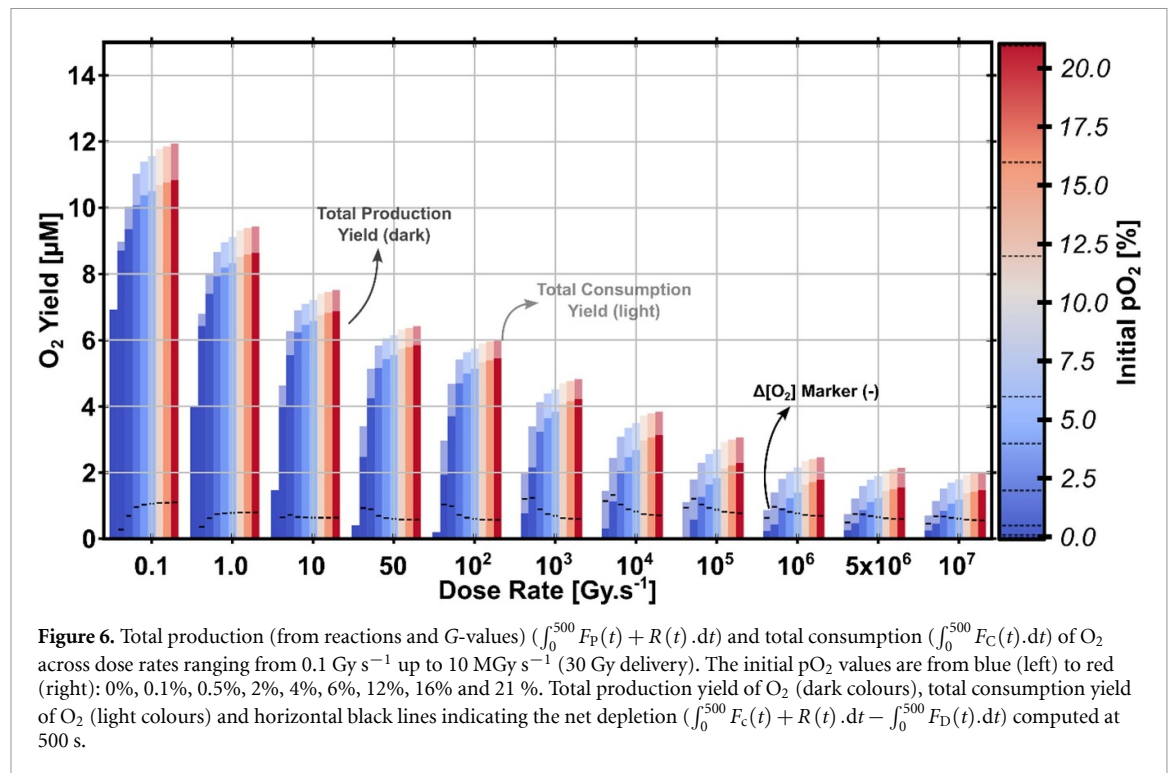


Figure 5. $\Delta[O_2]$ (expressed in %pO₂), computed at 500 s, as a function of initial oxygen content across dose rates ranging from 0.1 Gy.s⁻¹ up to 10 MGy.s⁻¹ (30 Gy delivery).

at pO₂ > 2%. The increase in dose rate reduces both consumption and productions yields in the same way. Therefore, it does not change the net behaviour.

For UHDR (≥ 100 Gy.s⁻¹): A global decrease is observed in the absolute values of both O₂ consumption and production yields. At high pO₂ levels, the change in $\Delta[O_2]$ is comparable to that observed under CDR conditions. However, at low pO₂ levels (pO₂ $\leq$ 2%), consumption yields exceed production



yields, resulting in a net increase in $\Delta[O_2]$ relative to CDR. Therefore, $\Delta[O_2]$ exhibits a maximum value within the low pO_2 range.

3.3. Hydrogen peroxide

Motivated by the work of other research groups (Montay-Gruel *et al* 2019, Blain *et al* 2022, Kacem *et al* 2022), the hydrogen peroxide production has been used as an indicator for cellular damage. H_2O_2 relative stability makes it suitable for experimental measurements and subsequent confirmation with computational methods. The concentration of hydrogen peroxide is extracted from the simulations results and its temporal evolution for the selected dose rates and initial oxygen content are plotted on figure 7.

The general evolution of H_2O_2 concentration is a steep increase during beam ON followed by a stable end value during beam OFF.

For hypoxia/physioxia (1%/5%) and CDR (figures 7(A) and (C)): The final H_2O_2 values are in the range $1.7\text{--}3.3 \mu\text{M}$. In the case of physioxia, values are more spread ($1.7\text{--}3.3 \mu\text{M}$) than in hypoxia ($2.0\text{--}2.7 \mu\text{M}$). For both cases an increase in dose rate reduces the final value of H_2O_2 .

For hypoxia/physioxia (1%/5%) and UHDR (figures 7(B) and (D)): Both hypoxia and physioxia show the same general evolution with the dose rate. However, in the case of hypoxia final values are 12%–26% higher than for physioxia. Moreover, unlike CDR cases, final values in hypoxia are more spread than in physioxia. In hypoxia, H_2O_2 values increase from $2.2 \mu\text{M}$ at $10^2 \text{ Gy}\cdot\text{s}^{-1}$ to $2.45 \mu\text{M}$ at $10^4 \text{ Gy}\cdot\text{s}^{-1}$ with a saturation already observed at $10^3 \text{ Gy}\cdot\text{s}^{-1}$. For dose rates $>10^4 \text{ Gy}\cdot\text{s}^{-1}$, the final H_2O_2 values steadily decrease down to $1.25 \mu\text{M}$ for $10^7 \text{ Gy}\cdot\text{s}^{-1}$. In physioxia, saturation of final values at $1.7\text{--}1.8 \mu\text{M}$ is already observed from $10^2 \text{ Gy}\cdot\text{s}^{-1}$. With increasing dose rate, the final H_2O_2 value steadily decreases down to $1.1 \mu\text{M}$ for $10^7 \text{ Gy}\cdot\text{s}^{-1}$.

As H_2O_2 concentration values are very stable after the beam delivery, the final values at 500 s are used to compare the production of this ROS over various dose rates and initial pO_2 , in a similar fashion as in figure 5. The net production of H_2O_2 is expressed with G -values unit and is computed by computing the slope of the linear fit of the H_2O_2 concentrations at end of simulation as a function of the dose.

There is a hot spot ($10^3 \text{ Gy}\cdot\text{s}^{-1}$, $pO_2 < 1\%$) on the H_2O_2 net production map of figure 8 for the same conditions as the hot spot observed for the $\Delta[O_2]$ map (see figure 5). To further examine and

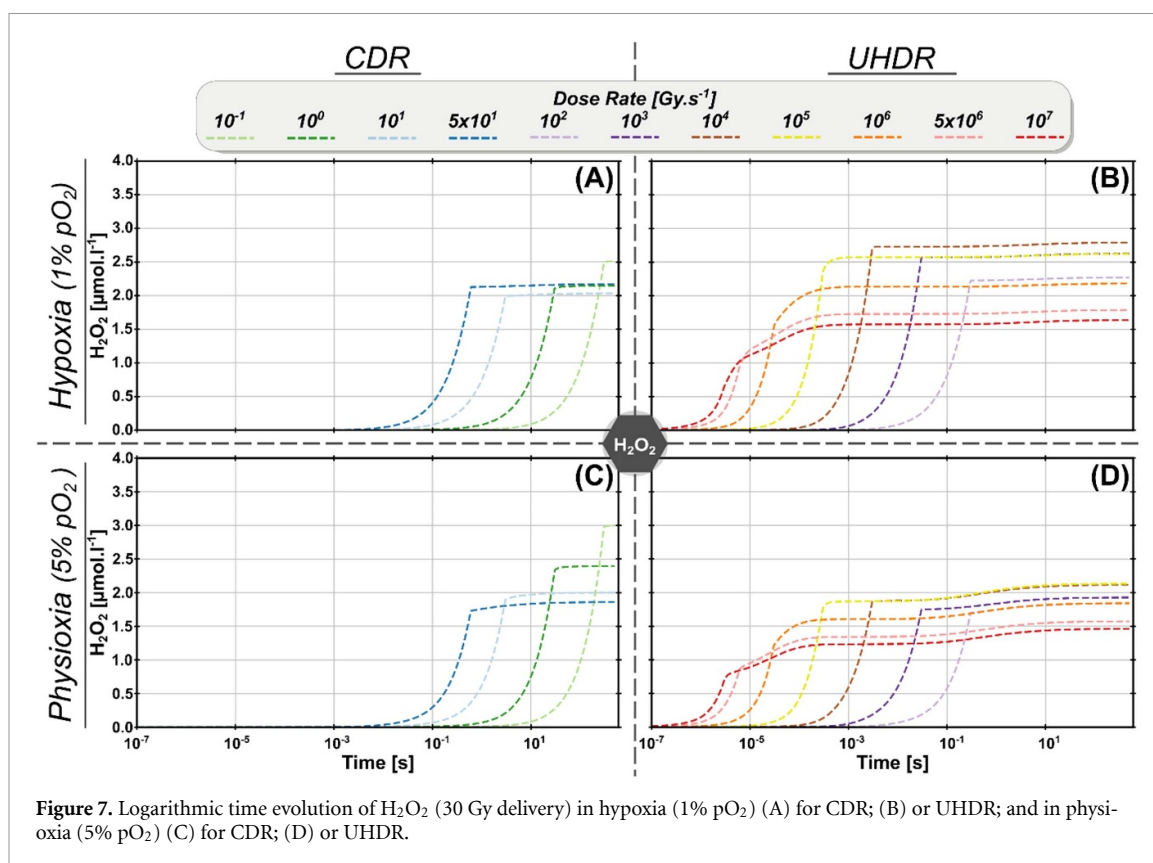


Figure 7. Logarithmic time evolution of H_2O_2 (30 Gy delivery) in hypoxia (1% pO_2) (A) for CDR; (B) or UHDR; and in physioxia (5% pO_2) (C) for CDR; (D) or UHDR.

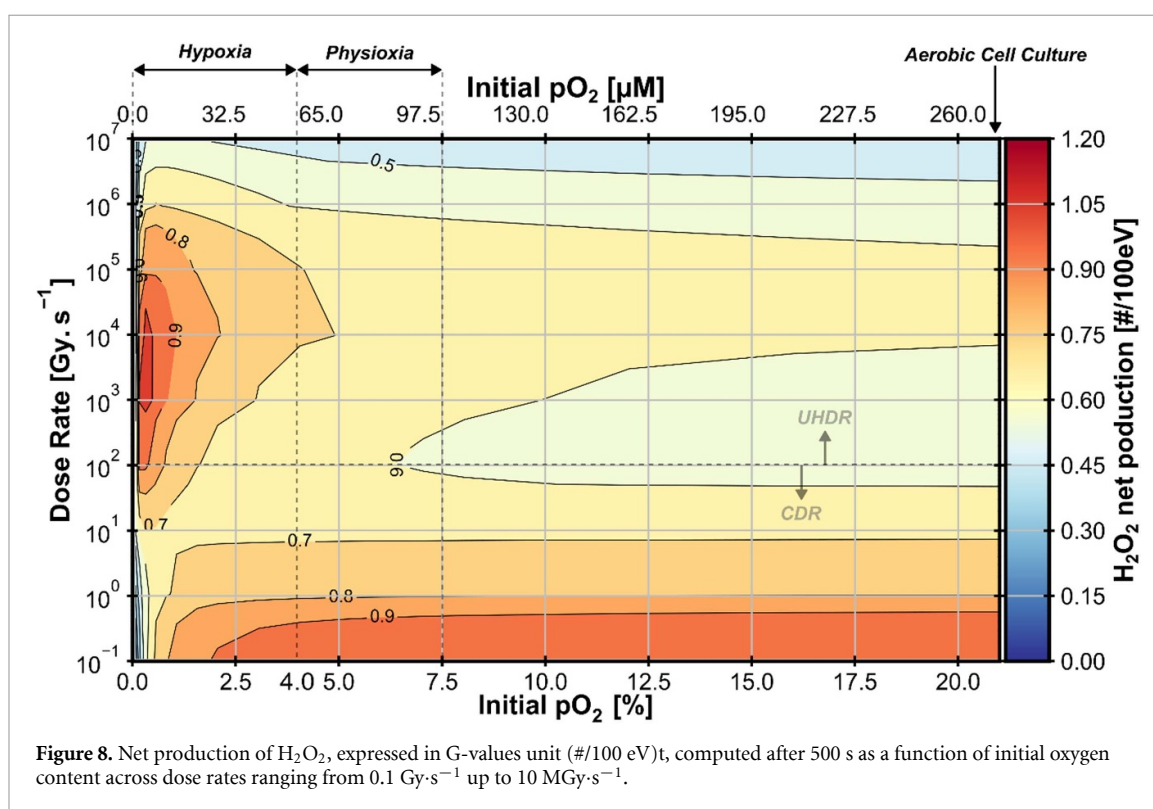
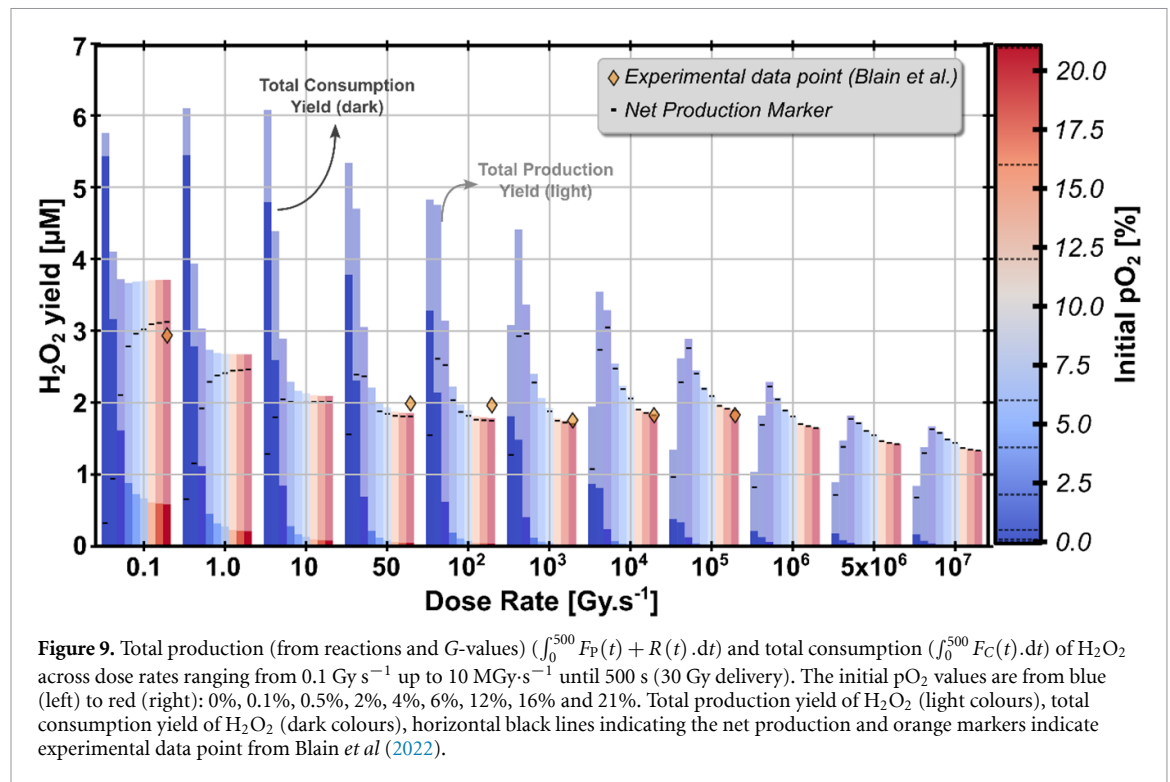


Figure 8. Net production of H_2O_2 , expressed in G-values unit ($\# / 100 \text{ eV}$)t, computed after 500 s as a function of initial oxygen content across dose rates ranging from 0.1 Gy.s^{-1} up to 10 MGy.s^{-1} .

understand the variations in hydrogen peroxide production, the time evolution of reaction rates producing or consuming H_2O_2 are analysed from simulation data. These time series are then integrated in the same fashion as figure 6 and presented on figure 9.



In figure 9, light colours represent gross production while dark colours represent consumption. As H_2O_2 is a primary product of water radiolysis, light colour bars always have a higher value. To ease the interpretation of the graph, the small horizontal black lines indicate the net production of H_2O_2 .

For CDR ($<100 \text{ Gy} \cdot \text{s}^{-1}$): initially ($pO_2 < 0.5\%$), H_2O_2 production and consumption yields have similar values, resulting to a net production $<2.3 \mu\text{M}$. As pO_2 increases, both yields decrease with consumption experiencing a more pronounced reduction. This differential decrease in consumption and production yields, leads to an overall increase in net H_2O_2 production, reaching $3.5 \mu\text{M}$ at the highest pO_2 levels. A general decrease with the dose rate is also observed for all yields.

For UHDR ($\geq 100 \text{ Gy} \cdot \text{s}^{-1}$): The general decrease continues for all yields but at a slower rate. For high pO_2 ($\geq 4\%$), the gross consumption is negligible, leaving the gross production dictating the net H_2O_2 production. The trend of H_2O_2 gross production with initial pO_2 changes from a monotonically decreasing function at CDR to a bell shape function with a maximum in the range 0.1%–2% pO_2 at UHDR. The net H_2O_2 production keeps decreasing with both the dose rate and the initial pO_2 level.

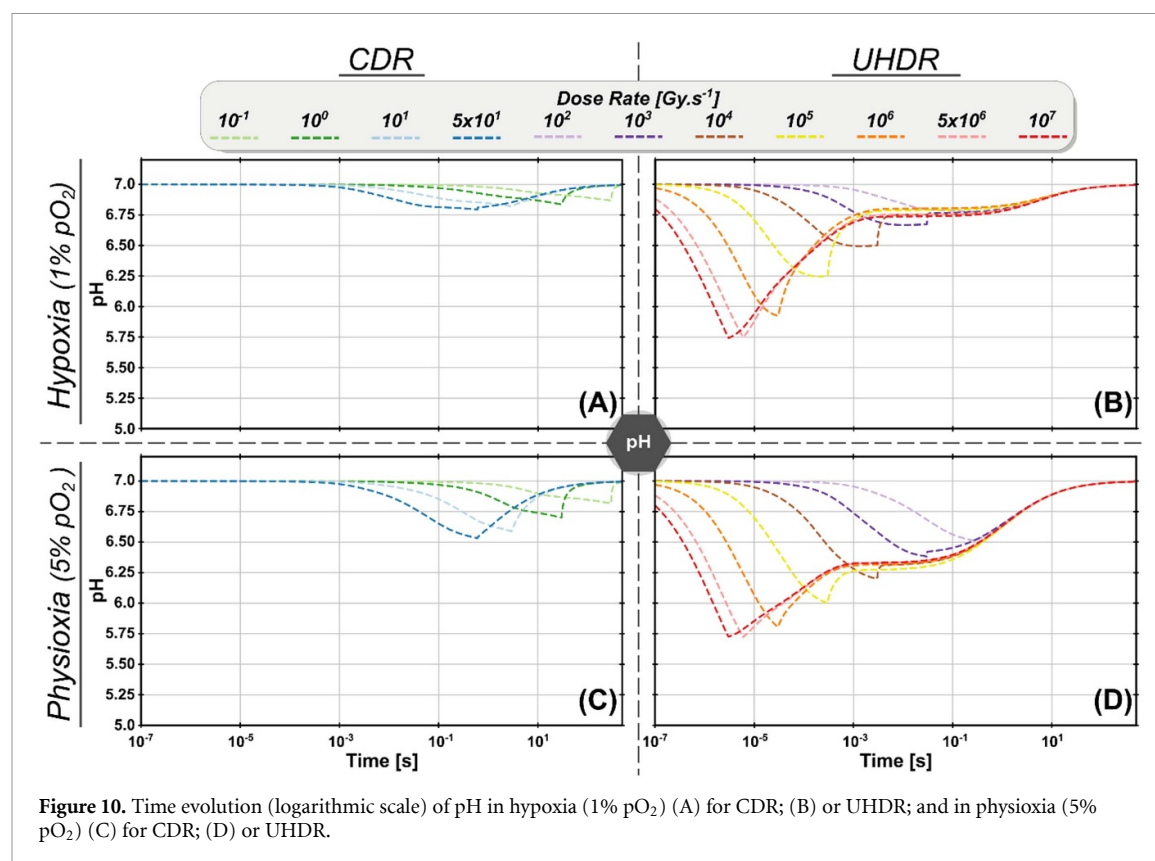
3.4. Early acidic spike

As a complete set of chemical reactions involving the H_3O^+ and OH^- ions are included in the model, it enables to track the time evolution of the acid–base equilibria and to compute the time evolution of pH via equation (5),

$$\text{pH} = -\log_{10} ([H_3O^+]). \quad (5)$$

Figure 10 displays the pH time evolution for the selected dose rates (see table 5) and hypoxia/physioxia.

The first observation of figure 10 is that the pH drops during irradiation, regardless of the dose rate or the initial oxygen content. When the beam is switched off, the pH value slightly rises back to its initial value of 7. Also, the higher the dose rate, the lower the minimal pH value. For CDR, all curves associated with physioxia (figure 10(C)) have lower minimum pH values than those associated with low pO_2 (figure 10(A)). The same applies for UHDR. These results also demonstrates that the initial amount of oxygen has a noticeable impact on pH recovery times: with increasing oxygen concentration, the time necessary for pH to recover escalates by multiple orders of magnitude, ranging from approximately 1 ms in hypoxia (figure 10(B)) to several seconds in physioxia (figure 10(D)).



4. Discussion

Wardman (Wardman 2022) outlines that chemical-kinetic modelling is a multi-step process: first, determine or propose a biological target or response of interest; second, develop a model consisting of reactions that may represent the chemical pathways to this response; third, gather rate constants and starting concentrations for all reactions and entities involved and solve the differential equations that describe the kinetics of the reaction set to estimate the concentration/time profiles of both reactants and products.

This study proposes an enhancement of a recognized method, previously established by other research groups (Labarbe *et al* 2020, Tan *et al* 2023), to explore the causes behind the emergence of the FLASH effect in biological systems. Compared to the other studies, the current model is focusing on the water-related chemistry initiated by the water radiolysis. Although it does not incorporate biological processes, this preliminary stage is essential for developing a more sophisticated model, as examining the effects of UHDR irradiation on pure water will serve as a foundational step. If differences from CDR exposure are observed, corresponding effects on biological responses can be expected. If no differences are found, it suggests that the FLASH mechanism must be sought elsewhere. The simplification of water radiolysis to focus only on primary reactions (i.e. those with the highest rate constants) is insufficient, as biological evidence suggests (Montay-Gruel *et al* 2019, Liljedahl *et al* 2022) that the effects of radiation on biological systems extend well beyond the duration of exposure. In this study, a model inspired by Labarbe *et al* (2020) was developed to study the effects of irradiation on pure water. The complexity of the chemical network is fully recognized, with the only simplification being the focus on the homogeneous chemistry phase, beginning 100 ns after the initial ionizations and excitations of water molecules. The effects of irradiation have been incorporated through widely accepted G-values (Boscolo *et al* 2020). The present model introduces several novelties: (i) a chemical network with high complexity, (ii) a comprehensive description of the acid-base equilibrium and (iii) a novel approach to account for the variation of G-values with dose rate. Specifically, the G-values are modified using a newly introduced coefficient, $G_F(D_R)$, whose evolution is determined based on experimental data and illustrated in figure 3. The behaviour of $G_F(D_R)$ follows a power law with negative exponent. For CDR regime, this coefficient is close to one whereas it reduces for higher dose rates. This trend is to be expected as G_{CDR} values are computed using single track simulations, which correspond to CDR conditions, resulting in relatively unchanged G-values. However, figure 3 highlights the need for additional experimental

data in the $1\text{--}40\text{ Gy}\cdot\text{s}^{-1}$ mean dose rate range to refine the fit of $G_F(D_R)$. The current lack of data in this range hinders a more accurate representation of $G_F(D_R)$. Even though the $G_F(D_R)$ coefficient has no direct physical meaning, it could be related to the intertrack interactions occurring at higher dose rates (Baikalov *et al* 2023). These interactions occur when reactive species from adjacent tracks overlap and react with each other, rather than with species within their own track. At UHDR, the high density of radicals and other reactive species can result in increased recombination reactions, reducing the overall yields of G -values.

In FLASH research, the production of hydrogen peroxide has been used as a surrogate to evaluate cellular damage. All available literature data are plotted together in figure 1. This study is questioning the main methods used to measure H_2O_2 as both Amplex Red and the Ghormley triiodide method produce diverging results. Several factors may explain these discrepancies, particularly the unreliability of the Amplex Red method under irradiation. First, the initial oxygen concentration in the sample, which is not always reported in the literature, can significantly alter the chemical reactions caused by irradiation and affect H_2O_2 levels (Petersson *et al* 2020, Scarmelotto *et al* 2024). Second, the Amplex Red method is complex and has several weaknesses when used in radiation experiments:

- **Enzyme and pH sensitivity:** The method relies on the HRP enzyme that is only effective within a specific pH range of 6–8 (Auriol *et al* 2006). Irradiation, especially at the UHDRs used in FLASH, is known to decrease the water's pH (Bepari *et al* 2023), which can inhibit the enzyme and lead to inaccurate H_2O_2 measurements.
- **Probe integrity:** If the Amplex Red probe and HRP enzyme are added before irradiation, they can be damaged by the radiation itself, compromising the results.
- **Chemical interference:** Under UHDR conditions, the fluorescent product of the Amplex Red reaction can be altered by other radiation-induced molecules (Dutton *et al* 1989). This interference can cause the signal to be either artificially low (underestimating H_2O_2) or high (overestimating H_2O_2).

Due to these significant limitations and the potential for unreliable results from the Amplex Red assay, this work exclusively uses the more robust Ghormley triiodide method for H_2O_2 measurements.

With the definition of G_F in figure 3(B), simulations at various initial oxygen levels are performed.

Figure 8 illustrates the effect of dose rate on hydrogen peroxide (H_2O_2) net production, revealing a clear differential response that mirrors the biological hallmarks of the FLASH effect. At CDR, H_2O_2 production increases with oxygen until it saturates. In contrast, at UHDR ($\geq 100\text{ Gy}\cdot\text{s}^{-1}$) irradiation shows a non-monotonic behaviour, with H_2O_2 production peaking in the hypoxic region (0%–1.5% pO_2) before decreasing at higher oxygen levels. This differential behaviour provides critical insights when mapped onto physiological conditions.

Firstly, in the hypoxic regime ($\sim 1\%$ pO_2), which is characteristic of the tumour microenvironment, the net production of H_2O_2 is nearly identical for both CDR and UHDR. Since H_2O_2 is a major contributor to radiation-induced cytotoxicity, this result provides a chemical basis for the iso-tumour control probability observed with FLASH.

Secondly, and perhaps more importantly, at higher oxygen levels representative of healthy tissue (physioxia, $\sim 5\%$ – 7% pO_2), the model predicts a significant reduction in H_2O_2 yield under UHDR compared to CDR. This aligns perfectly with the most crucial aspect of the FLASH effect: the sparing of normal tissue. While the long-standing oxygen depletion hypothesis attempted to explain this sparing, our simulations suggest a more intricate mechanism. The reduction in H_2O_2 is not merely a consequence of oxygen being unavailable but is rooted in a fundamental shift in how oxygen participates in the underlying water radiolysis chemistry at UHDR. As will be elaborated upon in the following section, our results demonstrate that oxygen plays a more subtle role, acting as a critical mediator in the radiolytic pathways governing H_2O_2 production. These combined observations are consistent with the core principles of FLASH evidence, assuming H_2O_2 is a reliable proxy for radiation induced cellular damage. To our knowledge, this is the first numerical model to reproduce this behaviour. The mechanism behind the decrease of H_2O_2 at UHDR is twofold. First, the increase of the dose rate has two concurrent initial effects (i) it reduces the total yield of primary irradiation products, and (ii) it increases their production rate. For instance, as shown in figure 3, an increasing dose rate, from 0.1 to $10^4\text{ Gy}\cdot\text{s}^{-1}$, results in a 77% reduction in G_F , while the delivery time decreases by five orders of magnitude. Second, the subsequent chemistry governing H_2O_2 production is primarily driven by the hydrated electrons, as previously suggested in (Blain *et al* 2022) but without the overall chemistry network involving acid–base equilibrium.

To elucidate the complex dynamics of H_2O_2 production and consumption, it is essential to examine the key reactions. Figure 11(A) provides a schematic representation of the main reactions associated with H_2O_2 and O_2 , featuring only those with the highest integral rates.

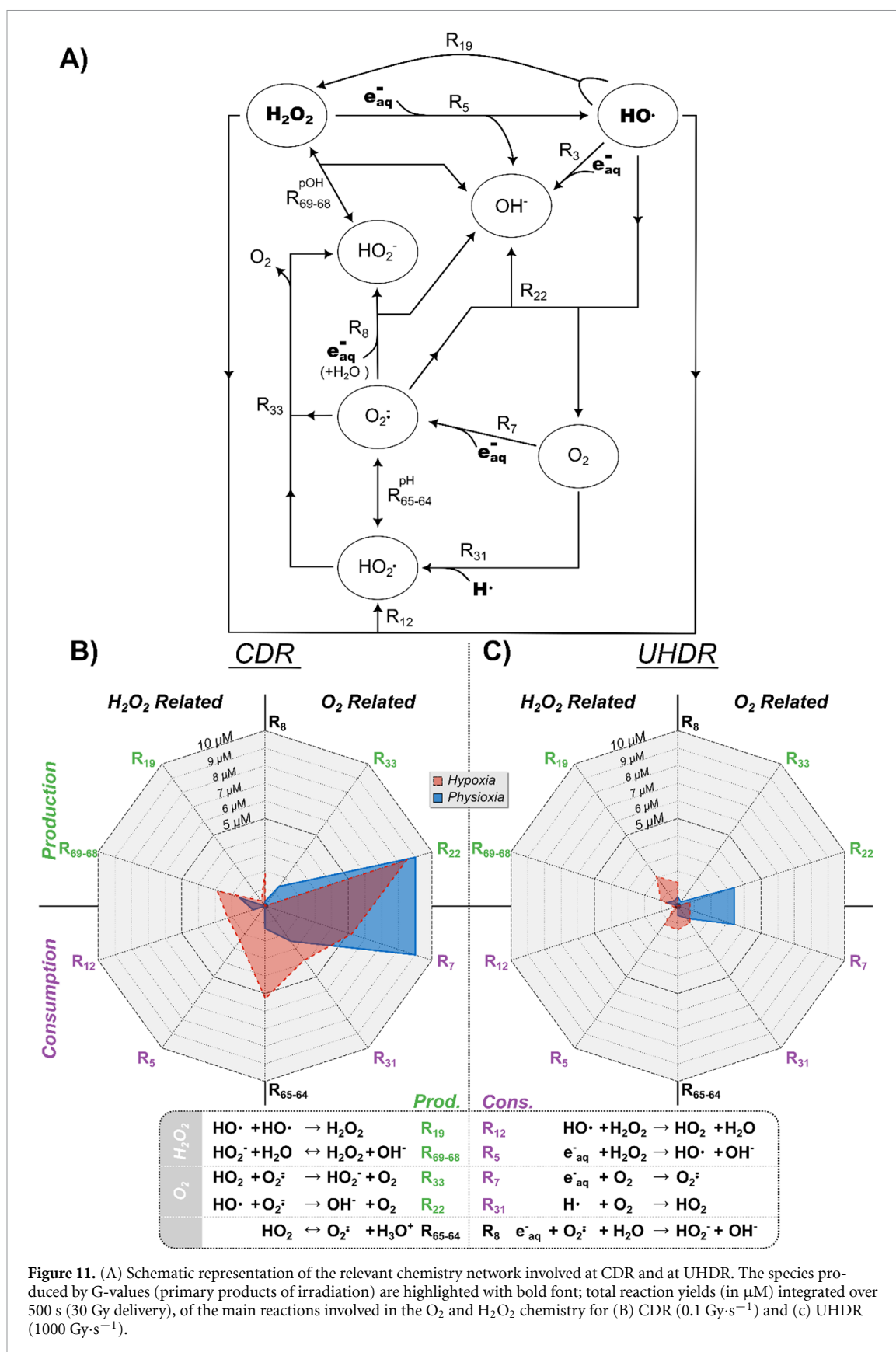


Figure 11(A) identifies three primary reactions responsible for H_2O_2 production: (i) radiolytic production based on G-values, (ii) self-recombination of $\text{HO}\cdot$ (R_{19}), and (iii) equilibrium reactions R_{68-69} which are pOH-dependent and link the concentrations of $\text{HO}_2\cdot$ and H_2O_2 . Conversely, H_2O_2 is consumed by two main reactions: (i) degradation by hydrated electrons (R_5) and (ii) reaction with $\text{HO}\cdot$ (R_{12}), yielding the $\text{HO}_2\cdot$ radical.

The diagram highlights hydrated electrons as the primary species responsible for H_2O_2 consumption, in agreement with previous findings (Iwamatsu *et al* 2018). However, their high reactivity and sensitivity to scavengers, such as O_2 , complicate the reaction scheme. To further analyse the system, figures 11(B) and (C) provide the integral reaction yields over the 500 s of simulation corresponding to a delivered dose of 30 Gy.

CDR in hypoxic conditions (figure 11(B)): the concentration of $\text{HO}\cdot$ is low, resulting in limited yields of R_3 and R_{19} . H_2O_2 , being a primary radiolysis product, accumulates as irradiation progresses, thereby enhancing the yields of R_5 and R_{12} , triggering a self-sustaining cycle. Specifically, R_{12} produces $\text{HO}_2\cdot$ which is then converted to $\text{O}_2^{\cdot-}$ through acid–base equilibrium reactions R_{64-65} ($\text{pK}_a = 4.9$), favouring $\text{O}_2^{\cdot-}$ production at physiological pH levels. The generated superoxide anions react with $\text{HO}\cdot$ (R_{22}) to form OH^- and O_2 , which in turn reacts with $\text{H}\cdot$ (produced as a primary product from the radiolysis of water and by the decomposition of H_2 with $\text{HO}\cdot$, not shown in the diagram) via R_{31} , to produce $\text{HO}_2\cdot$ perpetuating the cycle.

The radicals $\text{H}\cdot$ and $\text{HO}\cdot$ fuel this cycle, which persists as long as these primary products are available. Notably, R_{12} triggers the cycle even without initial oxygen (0% pO_2), highlighting oxygen's pivotal role, even when its net production is negligible. However, the cycle facilitates hydrated electrons consumption through reactions R_7 and R_8 . The main byproducts are O_2 , OH^- and $\text{HO}_2\cdot$ with the latter contributing to H_2O_2 production via R_{68-69} . The consumption of H_2O_2 by hydrated electrons (R_5) and hydroxyl radicals (R_{12}) counterbalances its gross production resulting in a negligible net production of H_2O_2 .

Similarly, $\Delta[\text{O}_2]$ is minimized due to the compensatory effect of reaction R_{22} on losses incurred by R_7 and R_{31} .

The high yield of reaction R_{22} combined with the yields from reactions R_5 and R_8 , generates a significant amount of OH^- ions, which contribute to gradual pH recovery by neutralizing hydronium ions (R_{59} , not shown in the diagram).

CDR in physioxic conditions (figure 11(B)): the hydrated electrons pool is quickly depleted by R_7 , significantly reducing H_2O_2 consumption via R_5 . The subsequent decrease in R_5 yield also reduces $\text{HO}\cdot$ production, which is then entirely consumed by R_{22} , effectively cancelling H_2O_2 consumption by $\text{HO}\cdot$ (R_{12}). Regarding H_2O_2 production, the reduced yield of R_8 (due to limited hydrated electrons availability) is partially offset by R_{33} , followed by R_{69-68} . As a result, the net production of H_2O_2 is higher compared to hypoxic conditions, primarily driven by R_{69-68} and the G -value, which are unaffected by pO_2 .

In this specific case, the self-sustaining cycle is partially bypassed by R_7 , which has a minimal impact on the overall chemistry since it directly branches to $\text{O}_2^{\cdot-}$. R_{22} maintains a high yield, regenerating consumed O_2 , although some O_2 consumed by R_7 is converted to H_2O_2 through R_{33} and R_{69-68} . This accounts for the saturation of $\Delta[\text{O}_2]$ observed in figure 6(B).

The reduced yields of R_5 and R_8 result in a lower total OH^- production, amplifying the acidic spike observed in figure 10(C).

UHDR in hypoxic conditions (figure 11 (C)): the reduction of G -values reduces the overall dynamic of the chemistry network. Still, in contrast to the CDR conditions, the higher concentration of hydrated electrons still enables all reactions in which they are involved (R_5 , R_7 , R_8 and R_3), although R_7 is less favoured due to the low O_2 concentration in hypoxic conditions. Under UHDR irradiation, the reaction rate of R_{12} ($2.92 \times 10^7 \text{ M}^{-1} \cdot \text{s}^{-1}$) is too slow to significantly consume H_2O_2 during the brief availability of hydrated electrons ($\sim$ beam ON time), making R_5 the dominant H_2O_2 consumption pathway. Additionally, under these specific conditions, most of the H_2O_2 is produced by R_{19} as the concentration of hydroxyl radicals is also enhanced at UHDR. The diminished yield of reaction R_{22} (due to the lack of $\text{HO}\cdot$ and $\text{O}_2^{\cdot-}$) disrupts the primary production route for O_2 . Consequently, when a small fraction of oxygen is added, the produced superoxide anion (by R_7 and R_{65-64}) is boosting the production of H_2O_2 via reactions R_8 and R_{69-68} . These factors explain why figure 8 shows a remarkable high net production yield of H_2O_2 at UHDR and low pO_2 (i.e. hypoxic conditions). Finally, from the perspective of OH^- production, the reduced production yield by reaction R_{22} is offset by the increased production yield of reaction R_3 . As a result, the drop in pH is tempered by OH^- (figure 10(B)),

although this drop is more pronounced than in the CDR regime due to an increase in H_3O^+ concentration as it is a primary product of irradiation.

UHDR in physioxenic conditions (figure 11(C)): similarly to CDR under physioxenic conditions, hydrated electrons are consumed by R_7 , which dramatically reduces the yields of reactions R_5 , R_3 and R_8 . The high yield of R_7 combined with the elevated concentration of $\text{HO}^\cdot$, stimulates reaction R_{22} . R_{12} is still negligible for the same reason as under UHDR-hypoxia. As a result, H_2O_2 consumption effectively ceases (see figure 9). Gross H_2O_2 production from reaction R_{19} is lower than under UHDR-hypoxia. Indeed, $\text{HO}^\cdot$ is consumed by R_{22} because of the increased oxygen concentration producing more O_2^- . Moreover, the production route via reactions R_{69-68} continues to be supported by reactions R_8 and R_{33} . Still, compared to CONV dose rates under physioxenic conditions, the net production yield of H_2O_2 is reduced. As the yield of reaction R_{22} is restored, the oxygen produced compensates for some of its consumption. Consequently, oxygen depletion is less significant than at UHDR under hypoxic conditions. Even though the yield of reaction R_{22} is higher compared to UHDR hypoxic conditions, it does not fully compensate for the substantial reduction in OH^- production resulting from decreased yields of reactions R_3 , R_5 and R_8 . This decrease leads to a lower total amount of generated OH^- and prolongs the recovery time from the acidic spike observed on figure 10(D).

The role of oxygen in FLASH-RT has been extensively investigated, emerging as a common thread in both *in vitro* and *in vivo* studies. *In-vitro* experiments have demonstrated that oxygen concentration significantly influences the FLASH effect. Notably, Vozenin *et al* (2019) found that aerobic cells (21% oxygen) did not exhibit the FLASH effect under UHDR irradiation. Similarly, Adrian *et al* (2019) showed that the differential response of prostate cancer cells to UHDR ($600 \text{ Gy}\cdot\text{s}^{-1}$) and CDR ($0.23 \text{ Gy}\cdot\text{s}^{-1}$) was dependent on oxygen concentration. Their results indicated that the difference in cell survival between UHDR and CDR, observed at physiologically relevant oxygen levels (1.6%–4.4% pO_2), diminished at elevated pO_2 ($>8.3\%$). *In-vivo* studies have also corroborated the oxygen dependence of the FLASH effect, with higher oxygen levels diminishing its tissue-sparing benefits (Zhang *et al* 2023, Tavakkoli *et al* 2024). Elevated oxygen concentrations have been shown to eliminate the FLASH effect, underscoring the critical role of oxygen in this phenomenon. These findings are consistent with the results of this study, assuming that the FLASH effect is not solely explained by direct damage to cells. Indeed, the net H_2O_2 production (figure 8) shows no differential effect between CDR and UHDR above 6% pO_2 . This differential effect, observed within a specific range of oxygen levels ($\text{pO}_2 \leq 2\%$), underscores the diversity of factors potentially involved in the FLASH effect and its oxygen dependence, as confirmed by multiple other studies (Adrian *et al* 2019, Montay-Gruel *et al* 2019, Petersson *et al* 2020).

While the oxygen depletion hypothesis has been a long-standing explanation for the FLASH effect, it has been largely disputed in the literature, with several studies arguing against it as the primary mechanism. Our findings support this nuanced understanding, highlighting instead the intricate role of oxygen in the radiolytic pathways. The observed correlation between oxygen depletion (figure 5) and the net production of H_2O_2 (figure 8) aligns with the recent experimental work of Sunnerberg *et al* (2023). In their study, which employed an electron beam ($\text{LET} \approx 0.2 \text{ keV}\cdot\mu\text{m}^{-1}$) compared to the proton beams modelled here ($\text{LET} = 1.14 \text{ keV}\cdot\mu\text{m}^{-1}$), both conditions remain within the low-LET regime where radiation-induced chemical damage mechanisms are assumed comparable (Lehnert 2007). In their study, the authors have measured both the production of H_2O_2 and the oxygen depletion in pure water samples containing 600 μM of bovine serum albumin after electron beam irradiation. Crucially, their study reports a statistical correlation between the amount of O_2 consumed and the amount of H_2O_2 produced per-Gy during irradiation. This direct quantitative link between oxygen consumption and H_2O_2 generation, strengthens the current findings by demonstrating a consistent relationship across different experimental modalities and the presence of a scavenger. The results presented here further elucidate this relationship between oxygen depletion and H_2O_2 production (in the case of pure water) through the chemistry network, highlighting the central role of oxygen in the production of H_2O_2 (figure 11). Specifically, oxygen is consumed to produce O_2^- which is either used to generate H_2O_2 or reconverted to O_2 by reaction with $\text{HO}^\cdot$ depending on dose rate and initial O_2 level. Thus, our study demonstrates that oxygen plays a more subtle role in the FLASH effect, not as a simple depletion mechanism, but as a critical component in the radiolytic pathways leading to H_2O_2 production.

The computed absolute values of oxygen depletion in this study ($\sim 0.03 \mu\text{M}\cdot\text{Gy}^{-1}$) are lower than those measured and predicted by other simulation codes (Pratx and Kapp 2019, Labarbe *et al* 2020, Tan *et al* 2023) ($0.2\text{--}0.7 \mu\text{M}\cdot\text{Gy}^{-1}$). However, assuming identical G_F for all primary product of irradiation is a strong simplification. A more accurate approach would be to make G_F species-dependent. Unfortunately, the current scientific literature lacks sufficient data to establish this dependence. Bepari *et al* (2023) have employed the MC multitrack irradiation model IONLYS-IRT (Meesungnoen and

Jay-Gerin 2010) to investigate the impact of dose rate and LET on the G -values associated with H_3O^+ . Their findings indicate that G -values decrease with increasing dose rate with a more pronounced effect for high LET ($>8 \text{ keV}\cdot\mu\text{m}^{-1}$). However, their study was limited to very high dose rates ($>10^6 \text{ Gy}\cdot\text{s}^{-1}$) and the underlying model implements an ‘instantaneous pulse’ model i.e. pulse duration assumed to be zero. This assumption differs from the present study and limits the possible comparison.

Finally, hydronium ions are directly created upon irradiation via G -value and are responsible for the acidic pH developed in the solution at early times (see figure 10). The temporary acidic spike observed at high dose rate is remarkable. With UHDR beams, the pH drops down to 5.25 and these acidic conditions persist for dozens to hundreds of milliseconds, varying with the starting level of oxygen. Only one other research group has forecasted this transient acidic peak through MC simulations (Sultana *et al* 2022, Bepari *et al* 2023). In their work, they reported a pH drop to 4.9 for a dose rate of about $10 \text{ MGy}\cdot\text{s}^{-1}$ with MC simulations of 300 MeV protons ($0.3 \text{ keV}\cdot\mu\text{m}^{-1}$) in pure unaerated water at 25°C . The present study predicts pH drop of 5.8 is consistent with these findings. When analysed within the FLASH framework, the pH behaviour presents a nuanced picture. The acidic peak is most pronounced under hypoxic conditions (the tumour iso-effect zone) and is progressively dampened as oxygen levels rise towards physioxia (the normal tissue sparing zone). Crucially, this transition is smooth and does not exhibit the ‘hotspot’ peak seen with H_2O_2 production under hypoxic UHDR. This suggests that while radiation-induced acidification may contribute to the overall biological effect, it does not, by itself, appear to drive the sharp differential between tumour control and normal tissue sparing. However, establishing a direct correlation between this prediction and cellular irradiation scenarios is not a simple task. This is due to the fact that it would necessitate the incorporation of buffers within the model. Despite the occurrence of reactions that counteract the increase in acidity, these reactions are not sufficiently rapid to mitigate the swift rise in H_3O^+ concentration. The incorporation of buffers, such as the carbonate–bicarbonate system (with reaction constant $\approx 10^{11} \text{ M}^{-1}\cdot\text{s}^{-1}$ (Cai *et al* 2017)), is anticipated to expedite a swift recovery. Nevertheless, pH disruption remains a likely outcome. The duration of pH recovery in biological systems, as well as the consequences, remain subjects of research. It is known that the regulation of the cellular microenvironment plays a role in the radio resistance of tumour cells (Arneth 2019, Rahman *et al* 2024) and that cancer cells exhibit a reduced pH in the inter-cellular environment, leading to the creation of an acidic microenvironment (Meng *et al* 2024). The effect of radiation-induced acidification on the microenvironment of either healthy or cancer cell may also hold key explanations in the elucidation of the FLASH effect.

The frontier between CDR regime and UHDR enabling FLASH effect is not clearly defined. As already mentioned, literature (Pratx and Kapp 2019, Romano *et al* 2022) generally suggests a threshold of about $100 \text{ Gy}\cdot\text{s}^{-1}$. In the light of the current results the net production of H_2O_2 might be proposed as a parameter to classify the two dose rate regimes. Indeed, hydrogen peroxide production is clearly delimiting two different behaviours on figure 9. At CDR, the net production yield (black markers) increases with the pO_2 and saturates while at UHDR the production yield increases, passes by a maximum and then diminishes. Based on these results, the frontier between the two irradiation regimes would lie in between 10 and $50 \text{ Gy}\cdot\text{s}^{-1}$, when considering low LET radiations ($<2 \text{ keV}\cdot\mu\text{m}^{-1}$). This value, coincide with the one suggested by Limoli and Vozenin in (Limoli and Vozenin 2023) where the limit is assumed to be within 40 – $100 \text{ Gy}\cdot\text{s}^{-1}$ depending on the involved tissue. Additionally, in figures 8, 40%–50% differential production of H_2O_2 , between anoxia and physioxia oxygen levels, is reached at a dose rate of $\sim 10^3 \text{ Gy}\cdot\text{s}^{-1}$, suggesting dose rate optimization for optimal FLASH effect. This optimization should consider the oxygen levels in both cancerous and healthy cells, as well as potentially other relevant parameters.

5. Conclusion

A comprehensive simulation model has been developed to investigate the homogeneous chemistry resulting from the irradiation of pure water under varying initial oxygen concentrations. This versatile model accommodates a wide range of dose rates, from 0 to $10^7 \text{ Gy}\cdot\text{s}^{-1}$, and allows for easy adaptation of the radiation source, requiring only the LET. Simulations can be executed over an extensive time scale, from milliseconds to minutes or even hours, depending on the study relevance. Notably, this innovative model incorporates a tentative model of dose rate dependent G -values and the kinetics of the acid–base equilibrium, enabling detailed analysis of pH dynamics occurring during irradiation. This work sheds the light on the distinct water-related chemistries involved in CDR and UHDR, as well as under hypoxic and physioxic oxygen levels. Understanding these differential impacts is crucial in the search for an explanation of the FLASH effect in biological systems.

Our results show that the complex chemistry of hydrogen peroxide production and consumption is significantly influenced by both dose rates and oxygen levels. Notably, this pure water model is the first to simultaneously predict (i) a decrease of hydrogen peroxide production in UHDR at physioxenic oxygen levels, consistent with experimental evidence from literature, (ii) behaviour consistent with the sparing effect for healthy cells in physioxenic conditions, and (iii) behaviour consistent with iso-tumour control probability for cancer cells hypoxic conditions.

Furthermore, this modelling tool enables us to probe and understand the complex and sometimes transient chemistry involved. This study highlights a dose rate-dependent acidic peak with the initial oxygen level significantly influencing both its amplitude and duration. Oxygen is once again identified as a key species, but with an indirect complex role in pH regulation and H_2O_2 production routes.

In the future, this numerical model might be improved by incorporating pH regulation processes of living cells to investigate the fate of the observed transient acidic peak generated by UHDR irradiations. Furthermore, adding Fenton reactions and enzymatic compounds such as catalase, superoxide dismutase, glutathione peroxidase, or peroxiredoxins could make the results more biologically relevant and improve the model's accuracy. Finally, a collaborative effort between experimentalists and modellers is required to elucidate the dose rate dependency of G-values and advance our understanding of UHDRs.

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary information files).

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